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REVIEW PAPER

UGV-UAV Integration Advancements for Coordinated Missions: A Review

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Abstract
The integration of Unmanned Ground Vehicles (UGVs) and Unmanned Aerial Vehicles (UAVs) represents a transformative leap in autonomous system coordination, combining ground mobility with aerial flexibility to tackle complex real-world missions. However, disparities in communication protocols, sensor interoperability, and autonomy frameworks pose persistent integration challenges. This review critically analyzes these gaps by synthesizing current literature and field practices. It introduces a novel Unified Robotic Protocol (URP) that enables real-time communication between ROS- and MAVLink-based systems, secured via blockchain technology. AI-driven swarm intelligence and modular payload systems are also proposed to enhance decentralized task allocation and operational adaptability. Key findings show significant improvements in system interoperability, mission coordination, and resilience under real-world constraints. This paper concludes with a roadmap for advancing secure, interoperable, and intelligent multi-agent systems.

Keywords Unmanned systems · Coordination · Interoperability · Autonomous operations · Sensor fusion

1 Introduction

Integrating UGVs with UAVs has been a huge step forward in unmanned vehicles over the last several years, where the merging of these two systems allow for the new system to tackle complicated problems in many different areas, such as surveillance, emergency response, and military operations [1]. In general, UGVs can carry huge cargo and could easily cross challenging terrains. These specific characteristics of UGVs enables the potential to enhance the performance of UAVs, which performs effectively in aerial surveillance, real-time data collection, and quick deployment [2, 3]. Despite of mechanical challenges in the development of unmanned platforms related to CAx [4, 5] due to wide range of carried out tasks, the operational synergy between these two unmanned vehicles has the potential to improve situational awareness, decision-making, and resource allocation when utilized concurrently [1, 2].

Recent studies have explored various approaches for enhancing unmanned systems. For instance, a study by Tang et al. introduces EN-MASCA, an enhanced multi-agent control algorithm for coordinating drone swarms in complex environments like durian orchards [6]. It incorporates deep reinforcement learning (DQN and PPO) and a virtual navigator model to enable real-time task adaptation, significantly improving path planning, obstacle avoidance, and swarm stability. The main advantage of real-time adaptation is its responsiveness to dynamic conditions, leading to higher patrol accuracy, formation consistency, and reduced flight deviations. Likewise, Alonso-Mora et al. proposes a method for reactive mission and motion planning for multi-robot systems in dynamic environments, with a focus on deadlock resolution and real-time collision avoidance [7]. It presents a

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correct-by-construction approach that synthesizes high-level task specifications into executable strategies, integrates with an optimization-based local motion planner, and generates environment assumptions when specifications are unrealizable. A key advantage is its ability to function without global knowledge, supporting decentralized obstacle avoidance and scalable operation with many dynamic agents. While these works provide valuable insights, they often lack a unified solution for cross-platform communication and robust data fusion. Moreover, current frameworks largely overlook blockchain-based security integration and decentralized task execution, which are essential for field deployments.

To address these gaps, our contributions include: (i) proposing a Unified Robotic Protocol (URP) that bridges the ROS and MAVLink ecosystems, (ii) introducing a secure, blockchain-verified data layer for tamper-proof communication, (iii) incorporating swarm intelligence for scalable, adaptive task planning, and (iv) validating these through real-world case studies. These contributions are positioned to overcome the fragmentation found in current UGV-UAV integration approaches.

Although there is clear potential for UGV-UAV integration, there are many obstacles that prevent these platforms from working together seamlessly. The intricacy of communication systems presents a significant challenge. Latency, signal interference, and protocol incompatibility between UGVs and UAVs impede the real-time transmission of essential data, representing prevalent issues in contemporary communication systems [2]. The sensor fusion process, essential for creating a unified comprehension of the operational environment, which is impeded by the variety of sensor systems employed by these platforms. Each system employs distinct data formats and processing standards [8].

This review examines these challenges and proposes solutions that utilize modern conceptual frameworks and emerging technologies. This review analyses how these advancements may enhance the interoperability and efficiency of integrated UGV-UAV systems. The construction of a URP is also included in this review, which is a hybrid communication framework that amalgamates existing protocols for UGVs and UAVs, hence enabling seamless data transmission and synchronization. This protocol addresses communication shortcomings and enhances security with blockchain technology, guaranteeing data integrity and secure multi-agent coordination.

This study analysis shows how modular payload systems for UGVs, and UAVs may adapt dynamically to changing mission requirements, boosting operational flexibility and mission success. UGVs may change payloads for supply delivery or surveillance, while UAVs can use solar panels to increase flight times. The review analysis will also examine how AI may improve mission planning by allocating workers dynamically and improve decentralized decision-making and resource efficiency by using artificial intelligence algorithms. More specifically, exploring swarm intelligence algorithms, which might be useful in complicated situations for coordinating many UGVs and UAVs.

This study combines contemporary research with a perspective that focused on the future, emphasizing emerging technologies that have the potential to alleviate existing issues. The importance of the solutions that are supplied may be attributed to the rising need for unmanned systems that are proficient, self-sufficient, and long-lasting, and that can function in challenging and unforeseen circumstances. This study aims to synthesize previous research with fresh suggested discoveries to help academics and developers better understand the possibilities for UGV-UAV integration. In addition, a strategy is proposed to tackle the current limitations in this dynamic field.

While UGV-UAV integration is a well-studied domain, this review introduces a unique contribution by proposing a Unified Robotic Protocol (URP) that not only bridges protocol incompatibilities but also integrates advanced technologies such as blockchain and swarm intelligence to improve system resilience, security, and autonomy. Compared to existing literature, this work distinguishes itself by addressing both communication interoperability and real-time multi-agent coordination in a hybrid robotic system under uncertain and constrained environments.

Notably, several advanced control strategies have been proposed in the literature. For instance, a quantized iterative learning control for communication-constrained systems was recently explored to tackle limitations in encoding/decoding mechanisms in networked control systems [9]. Furthermore, fault-tolerant mechanisms such as fuzzy wavelet neural adaptive control for UAVs operating under finite-time and self-triggered scenarios have demonstrated robust performance under high complexity environments [10]. Similarly in non-flying domains, like the robust control of reaction wheel-based robots, the emphasis on real-time adaptive control parallels the AI-driven strategies examined here for coordinated task execution [11].

While these studies present significant contributions in their respective domains, they focus primarily on individual vehicle optimization or control-specific innovations. This review, in contrast, shifts focus on multi-platform integration, examining how heterogeneous unmanned systems can be synchronized and secured under a unified operational framework. By drawing from advances in AI coordination, modular payloads, blockchain-enabled communication, and quantum encryption, this review offers a systems-level perspective aimed at advancing the interoperability and operational robustness of UGV-UAV teams in complex mission scenarios.

1.1 Problem Definition, Recent Advances, and Research Gaps

The integration of UGVs and UAVs for coordinated missions presents significant technical and operational challenges, stemming primarily from disparities in communication protocols, sensor systems, autonomy frameworks, and operational endurance. Current systems often lack interoperability, resulting in fragmented data exchange and inconsistent mission coordination. Moreover, cybersecurity vulnerabilities and inconsistent sensor fusion methods impede seamless integration, especially in real-time dynamic environments.

In response, recent advancements have introduced hybrid frameworks such as the Unified Robotic Protocol (URP), modular payload architectures, and AI-based coordination strategies including swarm intelligence. These solutions aim to bridge the technological gap between UGVs and UAVs by offering standardized communication, flexible mission adaptation, and decentralized task management.

Despite these innovations, several gaps persist. Firstly, there remains a lack of unified communication standards that operate reliably across both ground and aerial platforms under fluctuating environmental conditions. Secondly, real-world validation of AI-driven task allocation models is limited, and sensor fusion frameworks still struggle to provide consistent, coherent environmental maps. Lastly, emerging technologies like blockchain and quantum computing, though promising, are still in early stages of integration and lack comprehensive field validation.

The challenge in UAV-UGV integration primarily lies in the absence of a standardized framework for communication, coordination, and data fusion between these heterogeneous platforms. Current systems rely on distinct and often incompatible communication protocols, ROS (Robot Operating System) for UGVs and MAVLink (Micro Air Vehicle Link) for UAVs, which hinder real-time data exchange and synchronization. Moreover, sensor fusion complexities arising from varying sensor modalities (e.g., LiDAR for UGVs vs. optical and thermal imaging for UAVs) further exacerbate the issue by impeding a unified perception of the operational environment. Additionally, environmental and operational constraints, such as limited battery life for UAVs, terrain challenges for UGVs, and external interference in communication networks, restrict their effectiveness in real-world applications.

The challenges associated with UAV-UGV integration can be categorized into four key domains, which are communication and interoperability, sensor fusion and data integration, operational constraints, and autonomous task allocation and decision-making. In communication and interoperability, UAVs and UGVs utilize distinct communication protocols (ROS vs. MAVLink), leading to data format inconsistencies and transmission delays. In addition, signal interference and network latency impact real-time decision-making and system responsiveness, while the bandwidth limitations and cybersecurity vulnerabilities pose risks to mission integrity and operational security. On another note, in regard to sensor fusion and data integration, the fusion of heterogeneous sensor data (e.g., LiDAR for UGVs, optical/thermal cameras for UAVs) lacks standardization, leading to data synchronization issues. Furthermore, with high computational demands for processing multi-source data in real time, it also creates inefficiencies in perception and navigation. The disparities in sensor accuracy, resolution, and field of view also impedes the construction of a coherent operational map. In operational constraints on the other hand, UAVs suffer from limited flight endurance due to battery constraints, restricting their operational range, whereas UGVs encounter mobility challenges in rough terrain, limiting their effectiveness in certain environments. Adverse weather conditions also affect UAV stability and sensor accuracy, which in return will affect the mission reliability. Finally, the current issues in autonomous task allocation and decision-making are the lack of AI-driven coordination frameworks which would hinder adaptive mission planning and task distribution between UAVs and UGVs. Furthermore, as the centralized control mechanisms increase latency, it necessitates the adoption of swarm intelligence and decentralized decision-making strategies for dynamic mission execution. This review identifies and categorizes these challenges and explores how novel frameworks and technologies may be leveraged to address them, offering a consolidated roadmap for future research and development in autonomous multi-agent systems.

Additionally, the integration of UGV-UAV systems must consider the real-time operational challenges posed by highly coupled nonlinear dynamics, model uncertainties, external disturbances, and input constraints. These systems frequently operate in time-varying environments, where sensor feedback, environmental conditions, and mission parameters change unpredictably. Such dynamics can significantly degrade performance if not properly addressed. For instance, UGVs navigating uneven terrain while coordinating with UAVs in high-wind conditions may experience substantial nonlinear behavior and actuation limits. External disturbances such as gusts or uneven traction, combined with model inaccuracies in vehicle dynamics, necessitate robust control strategies that can adapt in real time. Furthermore, the presence of input constraints—such as limited actuation power or restricted communication bandwidth—further complicates seamless coordination. Addressing these issues requires advanced adaptive control algorithms, fault-tolerant designs, and uncertainty-aware task allocation frameworks, which will be further discussed in the following sections. Future research should continue to focus on quantifying and mitigating the impact of these real-world uncertainties to ensure resilient, mission-capable multi-agent autonomous systems.

1.2 Evaluation Metrics

To systematically analyse UAV-UGV integration strategies, a set of common evaluation metrics was defined, providing a structured framework for assessing existing methodologies. These metrics serve as quantifiable indicators for comparing UAV-UGV cooperative frameworks and identifying research gaps. The evaluation metric are as follows:

- Interoperability – This metric measures the compatibility of communication protocols, ensuring seamless data exchange across heterogeneous platforms.
- Real-time Communication – This will evaluate latency, bandwidth efficiency, and network robustness to support synchronized operations.
- Sensor Fusion Accuracy – This metric assesses the effectiveness of integrating multi-modal sensor data to generate a unified situational map.
- Autonomy and AI Integration – This examines the role of artificial intelligence in optimizing mission planning, obstacle avoidance, and decentralized decision-making.
- Operational Efficiency – This analyses energy consumption, endurance, and adaptability to dynamic environmental conditions.
- Cybersecurity & Data Integrity – This will evaluate encryption standards, authentication protocols, and blockchain security mechanisms to protect system integrity.

The selection of these metrics is grounded in their direct impact on UAV-UGV performance, ensuring that the review systematically evaluates technical feasibility, operational effectiveness, and security considerations.

1.3 Review Methodology and Contribution

This review utilizes a structured methodology grounded in six key evaluation metrics, interoperability, real-time communication, sensor fusion accuracy, autonomy and AI integration, operational efficiency, and cybersecurity. The assessment approach is comparative and evidence-based, focusing on analyzing the performance of UAV-UGV integration techniques across these criteria. A methodological flowchart is presented in Fig. 1 to guide the review process, from data collection to proposed framework formulation. Meanwhile, Table 1 shows the summary of the evaluation matrics.

The selection of these six-evaluation metrics; interoperability, real-time communication, sensor fusion accuracy, autonomy and AI integration, operational efficiency,

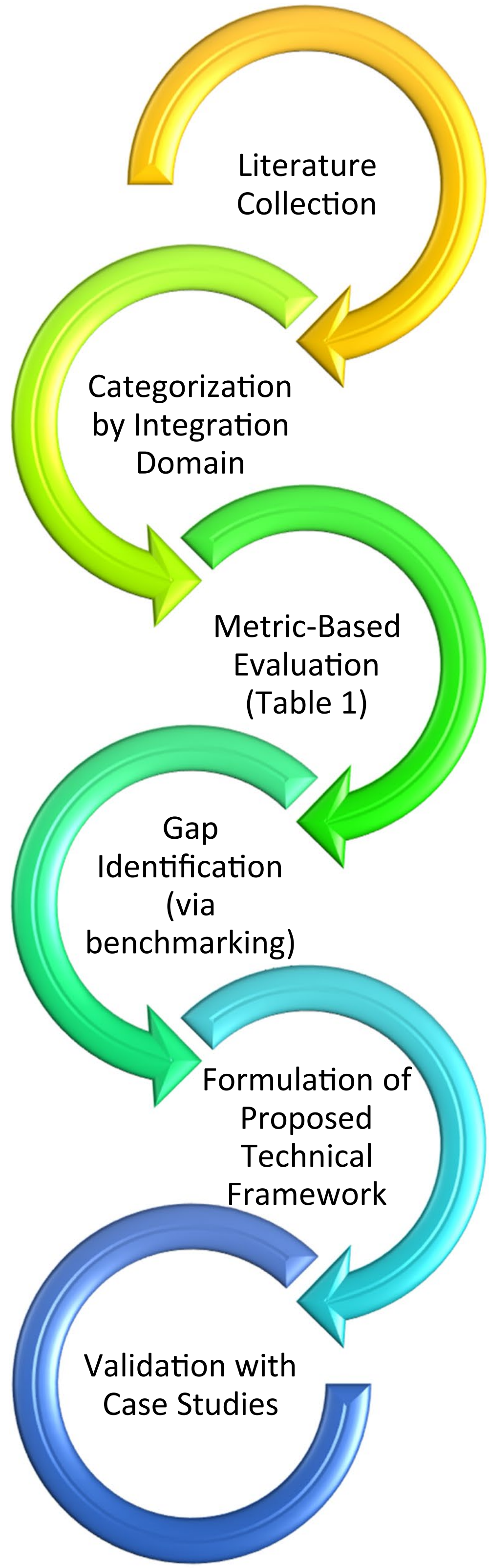


Fig. 1 Review methodology flowchart

Table 1 Evaluation metrics summary

Metric	Purpose	Evaluation tools
Interoperability	Assess cross-platform compatibility	Protocol comparisons
Real-time Communication	Measure latency and data throughput	Network tests
Sensor Fusion Accuracy	Validate environmental modeling	Sensor alignment algorithms
Autonomy & AI	Task efficiency & decision-making	Swarm AI flowcharts
Operational Efficiency	Energy, duration, terrain handling	UAV/UGV systems
Cybersecurity	Data integrity & resilience	Encryption frameworks

and cybersecurity, was strategically aligned with the core challenges identified in UGV-UAV integration. Interoperability directly addresses the incompatibility between communication protocols like ROS and MAVLink. Real-time communication is crucial in latency-sensitive applications such as search and rescue or military operations, where decision-making must occur instantaneously. Sensor fusion accuracy is necessary due to the heterogeneity of sensors (LiDAR, thermal, optical) and their role in building a unified environmental model. The autonomy and AI integration metric is critical in evaluating how well systems can perform decentralized task allocation using intelligent agents. Operational efficiency relates to power consumption, payload adaptability, and endurance, which are all essential for mission sustainability. Finally, cybersecurity and data integrity are essential to safeguard these autonomous systems against jamming, spoofing, and unauthorized access, particularly in adversarial environments. Each metric was chosen for its relevance to real-world deployment constraints and its role in advancing reliable multi-agent collaboration.

Through this comparative assessment, the study aims to:

i) Identify key limitations in current UAV-UGV coordination frameworks.
ii) Evaluate the effectiveness of emerging technologies in addressing interoperability, autonomy, and security concerns.
iii) Propose an optimized methodology to enhance UAV-UGV collaboration for real-world applications.

This review's main contributions are:

i) An in-depth analysis of UGV-UAV integration across technical, operational, and security dimensions.
ii) A proposed hybrid framework (URP) combining AI, blockchain, and cross-protocol communication.
iii) Case-driven insight on how to transition from theory to field applications, enhancing the real-world feasibility of coordinated unmanned systems.

1.4 Proposed Technical Framework for Improved UAV-UGV Integration

To overcome current challenges in interoperability, coordination, and secure data exchange, this review proposes a multi-layered technical framework (Fig. 2) that synthesizes modern innovations in robotic communication, AI coordination, and cybersecurity.

By integrating these advancements, this review aims to propose a scalable, secure, and intelligent framework for

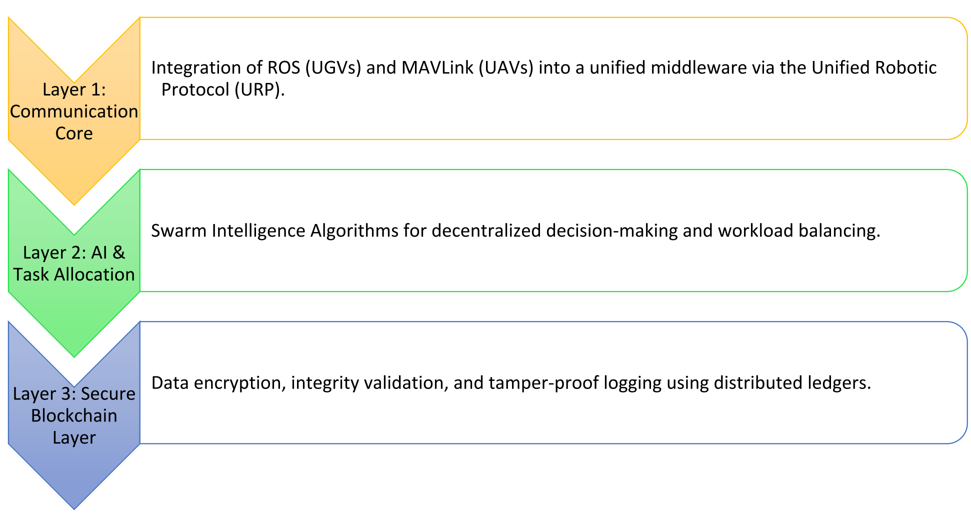


Fig. 2 Technical framework overview

UAV-UGV cooperation, fostering innovation in autonomous multi-agent systems. Thus, this review is divided into several sections to address the current issues and propose possible solutions that could address these issues. Section 2 presents the key challenges in integrating UGVs and UAVs for coordinated missions. It identifies and categorizes the major obstacles in their cooperation, primarily focusing on communication issues, cybersecurity vulnerabilities, sensor fusion, operational constraints, and real-time uncertainties. Section 3 focuses on novel solutions proposal to overcome the challenges of UAV-UGV cooperation. It introduces key advancements in communication protocols, AI-based task allocation, and modular system designs, providing a structured approach to enhance interoperability, efficiency, and autonomy. Section 4 explores the practical applications of UGV-UAV integration, highlighting case studies from military operations, disaster response, and surveillance. The section demonstrates how these autonomous systems complement each other in real-world missions, providing enhanced situational awareness, reduced human risk, and improved operational efficiency. Section 5 discusses emerging technologies that could enhance UAV-UGV cooperation, focusing on improving security, computational capabilities, and operational efficiency. It outlines three key areas of innovation which are blockchain for secure communication, quantum computing for mission optimization, and advancements in battery technologies. Section 6 summarizes the results and key findings found in this review. Finally, Sect. 6 concludes with future research directions, highlighting the key challenges and proposed solutions for UAV-UGV integration.

2 Challenges and Technical Barriers in UGV-UAV Integration

2.1 Communication Issues

A major obstacle in UGV-UAV integration lies in the differing communication systems and protocol standards each platform uses. UGVs typically rely on ground-based communication like Wi-Fi or cellular networks, while UAVs often use satellite communications or air-to-ground-specific technologies [12, 13]. These distinct protocols can create bottlenecks, packet loss, or complete failures in data flow. UGVs frequently use ROS (Robot Operating System), which is suited for processing complex sensor inputs such as LiDAR and stereo vision [14]. However, ROS often struggles with latency and scalability when scaled to aerial platforms. Conversely, UAVs rely on MAVLink, a lightweight, real-time telemetry protocol optimized for UAV mobility but inadequate for managing the heavy data loads required by UGV systems [15, 16].

The incompatibility between ROS and MAVLink is further exacerbated by environmental variables. UAVs generally utilize high-frequency bands like 2.4 GHz or 5 GHz, ideal for open-air operation but unsuitable for UGVs navigating obstructed terrain. UGVs benefit from sub-1 GHz bands, which offer greater penetration at the cost of lower data rates [17]. The need for seamless communication in dynamic, real-time operations—such as live video feeds and telemetry exchange in search-and-rescue missions—necessitates standardized, adaptive communication protocols. To address these issues, the Unified Robotic Protocol (URP) was developed, encapsulating both ROS and MAVLink into a unified format while using blockchain for secure, tamper-proof data transmission [18].

Another challenge arises in mesh network designs, which differ based on the vehicle type. UAVs often operate within hierarchical or hybrid mesh networks that use aerial nodes to extend communication range and enable line-of-sight connectivity [19]. UGVs, limited by terrestrial obstacles, favor decentralized flat topologies that offer resilience against node failures but at the cost of increased latency [20]. A hybrid mesh architecture has been proposed for balancing these strengths, combining UAV mobility with UGV stability for robust, large-scale operations [21]. Nevertheless, latency issues and communication loss remain significant risks, particularly when bandwidth-heavy data like video must be transmitted across varied platforms. Addressing these concerns may involve edge computing and optimized data transmission protocols.

Lastly, environmental factors introduce further complexity. UAVs require low-latency links for high-speed maneuvers and precise aerial control [22], while UGVs, which are more tolerant to delays, prioritize stable and durable connections for tasks like mapping or environmental monitoring [23]. The disparity in latency tolerance becomes critical when immediate decision-making is needed. Signal interference (whether from dense urban infrastructure or long-range rural transmission) further disrupts UAV-UGV collaboration [24–26]. As traditional networks fall short of handling these demands, software-defined networking (SDN), middleware like ROS2 DDS, and AI-driven network management are emerging as viable solutions to enhance interoperability and maintain reliable communication in complex environments [27, 28]. Figure 3 illustrates the communication challenges of UGVs and UAVs in their operational context.

There is a considerable relationship between the operational context and the intensity of these communication issues. Interference and protocol incompatibility are difficulties that are made worse in urban contexts, which are defined by dense infrastructure. Rural regions, on the other hand, are plagued by long-distance delays and restricted access to high-bandwidth networks, even though they have fewer physical impediments. These contextual differences

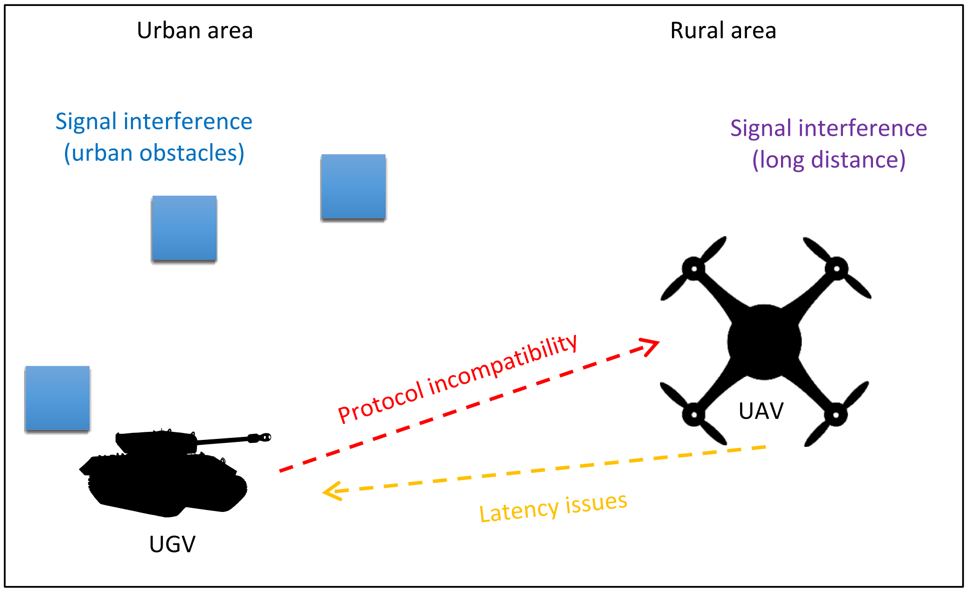


Fig. 3 Communication issues of UGV and UAV

emphasize the importance of adapting solutions to match the specific requirements of each site.

While hardware disparities such as differing frequency bands and transmission ranges contribute to communication inefficiencies, recent research emphasizes software-defined solutions to bridge these gaps. One promising direction is the adoption of Software-Defined Networking (SDN) to enable dynamic network configuration and centralized control over data flow between heterogeneous platforms. SDN facilitates seamless reconfiguration of routing paths and protocol translations, improving bandwidth utilization and reducing latency [27]. Additionally, ROS-MAVLink bridging middleware, such as MAVROS and ROS2 DDS (Data Distribution Service) implementations, allow protocol translation and unified data handling between aerial and ground platforms [28]. These tools abstract the underlying communication protocols, allowing UAVs and UGVs to share messages in real-time through standardized message formats.

On the sensor side, interoperability is increasingly addressed through middleware frameworks that support ontology-based sensor fusion and semantic data models. These systems enable sensor data from disparate sources (such as LiDAR, optical, and thermal sensors) to be semantically aligned, annotated, and integrated into a coherent understanding of the operational environment. Furthermore, multi-sensor data fusion libraries like Kalman filters, Bayesian networks, and deep learning-based sensor mapping are being adopted to reconcile resolution, scale, and temporal inconsistencies across platforms. These software-layer innovations represent a shift from rigid, hardware-dependent systems to more flexible and adaptive architectures that enhance the real-time interoperability of UGV-UAV systems.

It is essential to have procedures that are scalable, adaptable, and robust to address the connection challenges that arise between UGVs and UAVs. In order to realize the maximum potential of autonomous systems across a multitude of domains, it is important to address these issues. Research and development activities are primarily focused on the development of standardized communication protocols [29], the integration of advanced signal processing methods [30], and the application of artificial intelligence to adaptive network management [31]. The efficiency and reliability of cooperation between UGVs and UAVs can be enhanced by hybrid communication infrastructures that incorporate terrestrial, satellite, and relay-based networks.

The integration of UAVs with UGVs has significant potential for enhancing operational efficiency and effectiveness in challenging situations. Nevertheless, the challenges of communication, including signal interference, latency, and protocol incompatibility, present significant obstacles to the successful implementation of a fully integrated partnership. By addressing these challenges through system-level optimization and technical innovation, it is possible to facilitate more effective and dependable communication between these autonomous platforms, thereby paving the way for their widespread implementation in missions of particular significance.

2.2 Cybersecurity Challenges in UGV-UAV Cooperation

The coordinated operation of UGVs and UAVs introduces substantial cybersecurity vulnerabilities, primarily due to their reliance on wireless communication and autonomous systems. A critical issue is signal jamming, wherein adversaries deliberately overwhelm communication frequencies to disrupt data exchange [32]. Such interference can compromise mission-critical operations, causing delays or complete communication failure. This vulnerability is especially pronounced in urban environments or contested zones, where maintaining uninterrupted connectivity is paramount.

In addition to jamming, the risk of unauthorized access and hostile control of UGV-UAV systems poses significant threats [33]. Cyberattacks targeting unsecured communication protocols or exploiting weak encryption standards can allow adversaries to intercept sensitive information or take control of vehicles. In military contexts, this could lead to the diversion of assets, the spread of misinformation, or direct sabotage of critical operations. Civilian applications, such as disaster response, are not immune to these risks, as compromised systems could exacerbate the chaos in already volatile situations.

Another prevalent threat is spoofing, a technique where attackers manipulate navigation or control signals [34]. For instance, UAVs may be misled into restricted or dangerous airspace through GPS spoofing, while UGVs could be redirected into obstacles or areas with hostile conditions. These attacks degrade situational awareness, disrupt coordinated manoeuvres, and reduce the operational effectiveness of integrated systems.

To address these cybersecurity challenges, a multi-faceted strategy is required. Advanced encryption techniques must be integrated into communication protocols, ensuring secure data exchange and robust authentication mechanisms [35]. The adoption of blockchain technology offers a promising solution by providing a tamper-proof and decentralized framework for data management [36]. This ensures the integrity of transmitted information while enhancing trust among the system's agents. Additional measures, such as frequency-hopping spread spectrum (FHSS) communication, can mitigate jamming attempts by dynamically altering frequencies during operation, making interference significantly more challenging [37]. Moreover, quantum key distribution (QKD) is emerging as a ground-breaking method for achieving unbreakable encryption, effectively neutralizing the threat of signal interception or tampering [38].

Table 2 presents a comparative evaluation of various cybersecurity measures that address the vulnerabilities inherent in UGV-UAV systems. These measures are analysed based on their effectiveness against potential threats, implementation complexity, and associated costs. The comparison offers critical insights into selecting appropriate security solutions for diverse operational scenarios.

Basic encryption represents the foundational layer of cybersecurity employed across many autonomous systems [39]. Its primary appeal lies in its low cost and simplicity, making it a practical choice for civilian and less resource-intensive applications. However, as adversaries develop increasingly sophisticated attack techniques, such as advanced decryption methods and brute-force attacks, basic encryption has become less effective against high-level threats. This limits its applicability in scenarios where the integrity of mission-critical data or vehicle control is paramount, such as military operations or high-risk disaster response efforts.

Frequency Hopping Spread Spectrum (FHSS) communication, on the other hand, offers a robust defence against jamming attacks [41]. By dynamically altering communication frequencies during operation, FHSS effectively complicates the adversary's ability to disrupt transmissions. Its proven efficacy against signal interference makes it particularly valuable in contested environments where stable communication channels are crucial. Nevertheless, implementing FHSS requires moderate technical expertise and infrastructure, and the associated costs may exceed the budget constraints of smaller-scale or resource-limited deployments.

Blockchain technology stands out as a highly advanced solution due to its tamper-proof and decentralized nature [43]. The architecture of blockchain ensures that data exchanges between UGVs and UAVs are immutable and transparent, significantly enhancing trust and data integrity in complex operations. Blockchain is particularly effective in mitigating threats such as data manipulation, unauthorized access, and system spoofing. However, its adoption is not without challenges. The integration of blockchain into existing systems demands significant computational resources, expertise, and investment, which can pose barriers to widespread implementation. For this reason, blockchain is currently most applicable in high-stakes applications where security is non-negotiable, such as defence and sensitive industrial operations.

Table 2 Comparison of cybersecurity solutions

Solution	Effectiveness against threats	Implementation complexity	Cost
Basic Encryption	Moderate (vulnerable to advanced hacks) [39]	Low [40]	Low [40]
Frequency Hopping (FHSS)	High (against jamming) [41]	Medium [42]	Medium [42]
Blockchain	Very High (tamper-proof) [43]	High [44]	High [44]
Quantum Key Distribution	Unbreakable [45]	Very High [46]	Very High [46]

Quantum Key Distribution (QKD) emerges as the pinnacle of cybersecurity measures, offering encryption that is theoretically unbreakable [45]. By leveraging the principles of quantum mechanics, QKD ensures secure communication channels resistant to even the most advanced computational attacks. Its unparalleled effectiveness makes it an ideal solution for missions requiring absolute data confidentiality, such as classified military operations or critical infrastructure monitoring. However, QKD's implementation complexity is considerable, necessitating specialized hardware, extensive expertise, and significant financial investment. As a result, its current use remains limited to highly specialized fields with substantial resources and high-security priorities.

Table 2 emphasizes that no single cybersecurity measure is universally optimal for all scenarios. Basic encryption may suffice for applications with minimal security requirements, while FHSS provides a cost-effective yet robust option for environments prone to signal interference. Blockchain and QKD, despite their higher costs and complexity, offer transformative potential for safeguarding autonomous systems in high-stakes missions where security cannot be compromised. This underscores the need for a tailored approach to cybersecurity, where the chosen measures align with the specific operational demands, threat levels, and resource constraints of the deployment environment.

As technology advances, it is anticipated that the cost and complexity associated with innovative solutions like blockchain and QKD will decrease, broadening their accessibility and applicability across a wider range of use cases. In the meantime, decision-makers must carefully weigh the trade-offs between cost, complexity, and security to ensure the effective protection of UGV-UAV systems while maximizing operational efficiency.

Beyond technological innovations, a comprehensive approach involving regular security audits, real-time intrusion detection systems, and robust fail-safe mechanisms is essential [47]. These measures not only protect against known threats but also prepare systems to respond adaptively to emerging cyber risks. By implementing these strategies, the integration of UGVs and UAVs can achieve greater operational resilience, ensuring their effectiveness in critical applications ranging from military missions to disaster response. The continual evolution of cybersecurity measures will remain pivotal as these autonomous systems become increasingly central to modern operations.

2.3 Sensor Fusion and Data Integration

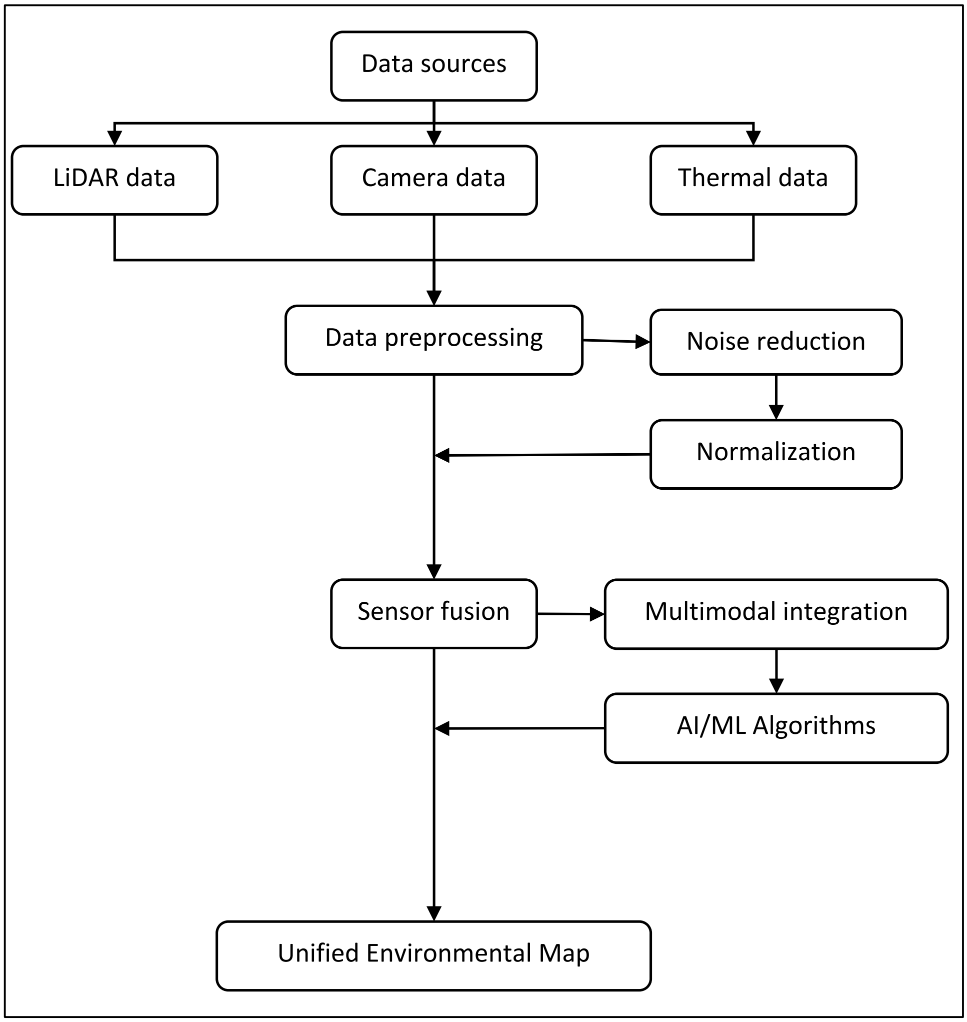
Sensor fusion plays a crucial role in creating a unified understanding of the environment in which UGVs and UAVs operate. However, achieving seamless integration of data from disparate sensors remains a significant challenge. UGVs and UAVs often employ different types of sensors, such as LiDAR, radar, cameras, and thermal sensors, each with unique data formats, resolutions, and processing requirements [2]. For instance, LiDAR sensors used in UGVs provide high-resolution 3D data essential for ground navigation [48], while UAVs often rely on cameras or thermal sensors to capture aerial imagery and detect heat signatures [49]. The fusion of these data streams is necessary to construct a comprehensive, real-time map of the operational environment, which can be critical for tasks like obstacle avoidance, target identification, or mission planning. Figure 4 shows the flowchart of sensor fusion in UGVs and UAVs.

Sensor fusion refers to the process of combining data from multiple sensors to generate a cohesive and accurate representation of the environment. Each sensor type offers unique advantages. LiDAR Sensors provide detailed 3D point clouds, capturing geometric information such as object shapes and distances [50]. Cameras capture high-resolution RGB images that contribute colour and texture data, aiding in object recognition and classification [51]. Meanwhile, thermal sensors detect heat signatures, which are critical in low-visibility conditions such as darkness, fog, or smoke [52]. While each sensor has strengths, they also have limitations. For instance, LiDAR data lacks colour information, cameras can be impeded by poor lighting, and thermal sensors may have lower resolution. By fusing data from these sensors, the limitations of one can be compensated by the strengths of others, resulting in a more reliable and robust environmental map.

Another primary issue with sensor fusion lies in the different nature of the data sources. Variations in data formats and processing standards complicate the development of algorithms capable of merging these inputs into a cohesive model [53]. Furthermore, the computational demands associated with real-time sensor fusion are non-trivial. High-quality sensor data must be processed and integrated efficiently, with minimal delay, to ensure that the system can respond to changes in the environment in real-time. Discrepancies in sensor accuracy, resolution, and field of view further complicate the fusion process, making it difficult to generate an accurate, unified model of the surroundings [54].

Before sensor data can be integrated, it must undergo preprocessing to ensure compatibility [55]. This stage involves several key operations which are noise reduction and normalization. The raw sensor data often contains errors or irrelevant information due to environmental factors such as weather, reflections, or electromagnetic interference. In the noise reduction stage, preprocessing removes these anomalies, improving data reliability. Different sensors also produce data in diverse formats (e.g., point clouds, pixel grids, heat maps). Thus, in the normalization stage, it standardizes these formats to align scales, resolutions, and coordinate systems, ensuring that data from multiple sources can be meaningfully compared and integrated [56]. The preprocessing

Fig. 4 Process of sensor fusion



step is essential to transform raw, unstructured data into a form suitable for fusion. Without preprocessing, sensor fusion algorithms may yield inaccurate or incomplete results.

Once data is pre-processed, it undergoes sensor fusion. This process integrates data streams from various sensors into a unified representation. The complexity of sensor fusion lies in reconciling differences in sensor modalities such as multimodal integration and artificial intelligence (AI) [57]. Data from LiDAR, cameras, and thermal sensors must be aligned spatially and temporally. For example, a detected object's location in the LiDAR point cloud must correspond to its appearance in the camera's RGB image and thermal signature. Advanced algorithms play a pivotal role in resolving conflicts or gaps in data. For instance, if one sensor fails to detect an object due to occlusion or noise, AI can infer missing information based on patterns from other sensors. The output of this process is a seamless dataset that combines the spatial precision of LiDAR, the visual richness of cameras, and the thermal sensitivity of heat sensors.

The goal of sensor fusion is to produce a Unified Environmental Map, which serves as the system's understanding of the environment. This map combines the strengths of all sensors, offering 3D spatial awareness (a comprehensive layout of objects and terrain features derived from LiDAR data), visual and textural details (enhanced by the RGB images from cameras, enabling advanced object recognition), and heat signatures (highlighting areas of interest such as humans, animals, or heat-emitting machinery). This unified map is invaluable for decision-making in a variety of applications. In search-and-rescue missions, for instance, the map enables a UGV to navigate rubble while identifying heat signatures indicative of survivors. Similarly, in agricultural monitoring, UAVs can provide detailed insights into crop health by fusing thermal and visual data.

Despite its advantages, sensor fusion poses several challenges. One of the challenges is data synchronization [58]. Sensors often operate at different frequencies and resolutions, making it difficult to align data streams in real-time. Another problem with sensor fusion is the computational overhead. The integration of high-resolution data from multiple sensors demands significant computational resources, which can strain hardware systems. Finally, factors such as lighting, weather, and obstructions can affect sensor performance, complicating the fusion process. Addressing these challenges requires robust algorithms, efficient hardware, and adaptive systems capable of handling diverse operational scenarios.

Sensor fusion and data integration are indispensable for enabling UGVs and UAVs to operate effectively in dynamic environments. By combining the complementary strengths of LiDAR, cameras, and thermal sensors, these systems can overcome the limitations of individual sensors and achieve a holistic understanding of their surroundings. While challenges such as synchronization, computational complexity, and environmental variability persist, ongoing advancements in AI and sensor technology are steadily improving the reliability and efficiency of sensor fusion. As the foundation for autonomous decision-making, sensor fusion is a critical step toward the realization of fully autonomous systems in fields ranging from disaster response to precision agriculture.

2.4 Environmental and Operational Constraints

The performance of integrated UGV-UAV systems is heavily influenced by environmental factors, which can severely impact their operational effectiveness. UAVs, for example, are highly sensitive to adverse weather conditions such as high winds, heavy rain, or fog, which can destabilize flight and reduce sensor accuracy [59]. In challenging weather, the performance of UAVs is particularly compromised as they become more vulnerable to turbulence, which can affect both their flight stability and the reliability of onboard sensors. For instance, poor visibility in foggy conditions can reduce the effectiveness of optical and infrared sensors used for navigation and surveillance, limiting the UAV's ability to perform precise tasks.

UGVs, while less sensitive to weather, face their own set of environmental challenges. Rugged and uneven terrain, such as rocks, mud, sand, or snow, can hinder mobility and make it difficult for the UGV to maintain stability [60]. Moreover, UGVs are often deployed in environments where obstacles like debris or steep slopes are present, which may restrict their ability to navigate effectively. These terrain-related challenges require UGVs to be equipped with advanced mobility systems that can adapt to a wide range of surfaces and obstacles. The coordination between UGVs and UAVs in such environments becomes particularly complex, as each system must adapt to its respective environmental conditions while maintaining synchronization with the other platform.

Operational constraints, such as limited battery life, also affect both UGVs and UAVs [2]. UAVs typically have shorter operational ranges due to the energy demands of flight, while UGVs may face similar limitations when tasked with carrying heavy payloads or navigating energy-intensive terrains. The integration of these systems into long-duration missions requires the development of energy-efficient technologies and adaptive power management systems to extend operational times and reduce downtime.

Table 3 shows the comparison between UGVs and UAVs highlighting their complementary roles in addressing operational challenges in varying environments. UGVs excel in navigating ground-level obstacles, carrying heavy payloads, and maintaining longer operational durations in stable conditions, making them ideal for industrial and search-and-rescue missions in rugged terrains [69]. Conversely, UAVs are optimized for aerial tasks that require agility, rapid deployment, and access to hard-to-reach areas, despite their sensitivity to adverse weather and shorter operational times. Recent systematic reviews have identified critical control and optimization challenges in UAV operations, emphasizing the need for metaheuristic solutions to adapt to diverse environmental and operational constraints [70].

By understanding these constraints, operators can strategically deploy UGVs and UAVs in tandem, leveraging their strengths to overcome their limitations and enhance mission efficiency across diverse applications. This complementary approach underscores the importance of integrating these platforms in modern autonomous and semi-autonomous operations.

2.5 Real-Time Uncertainty

Real-time autonomous operations of UGV-UAV systems face a spectrum of uncertainties that significantly affect their performance, safety, and coordination. These uncertainties can be broadly categorized as:

i. Internal vs. External—Internal uncertainties originate within the system itself, such as actuator nonlinearities, sensor noise, battery degradation, or computational delays. External uncertainties arise from the environment, including wind gusts, terrain variability, dynamic obstacles, communication losses, or adversarial interference.
ii. Parametric vs. Non-Parametric—Parametric uncertainties involve variations in known model parameters (like vehicle mass, friction coefficients, or moment of inertia) whose exact values may not be known in real time. Non-parametric uncertainties are more complex and represent completely unknown or unmodeled dynamics (new terrain types, unforeseen payload shifts, or unpredictable human/animal motion in operational zones, etc.).

Table 3 Operational constraints of UGVs and UAVs

Aspect	UGVs	UAVs	Source
Terrain Challenges	- Struggles with rough, uneven, or muddy terrains - Susceptible to obstacles like rocks, trees, and slopes	- Terrain is irrelevant as UAVs operate above ground - May face difficulties in confined areas or dense forest canopies	[2]
Weather Sensitivity	- Less affected by weather conditions unless extreme (e.g., flooding, or heavy snow) - Ground visibility and traction are crucial	- Highly sensitive to wind, rain, and snow - Adverse weather impacts stability, flight safety, and sensor performance	[61, 62]
Energy Constraints	- Typically powered by batteries or fuel - Energy consumption increases on difficult terrain or steep inclines	- Limited battery life due to high energy consumption for propulsion and hovering - Solar-powered UAVs may extend operational time in sunny conditions	[13, 63]
Operational Duration	- Longer durations (several hours) on flat, stable terrain - Reduced duration on challenging terrain due to increased power demands	- Typically, shorter operational durations (30 min to a few hours) for battery-powered UAVs - Can be extended with tethered power systems or optimized flight paths	[64, 65]
Payload Capacity	- Higher payload capacity than UAVs - Can carry heavy equipment, sensors, or supplies	- Limited payload capacity due to weight restrictions - Affects flight time and agility	[66]
Environmental Conditions	- Performs well in most environments except extreme temperatures and flooding - Sand, mud, and snow can impede movement	- Vulnerable to adverse weather conditions, high altitudes, and electromagnetic interference - High humidity can affect sensitive electronics	[53, 67]
Navigation Constraints	- Requires accurate mapping to avoid obstacles - GPS signals can be obstructed in dense urban areas or forests	- Relies on clear GPS signals for navigation - May struggle in GPS-denied areas or regions with high signal interference	[68]

In real-time deployment, the structure and magnitude of these uncertainties are typically unknown, making model-based prediction and planning highly challenging. For instance, UAVs navigating urban canyons may face fluctuating wind vortexes and GPS signal reflections, while UGVs might encounter unseen debris, changing terrain traction, or sudden load changes due to payload drop-off. These challenges directly impact stability, path planning, and coordination with other agents.

Uncertainty plays a critical role in mission resilience. Systems that fail to account for uncertainties often show degraded behavior such as mission drift, excessive energy consumption, or collision risks. Therefore, managing uncertainty is not just a performance issue, where it's foundational to the safety and reliability of UGV-UAV collaboration. For instance, in disaster response or military operations, abrupt terrain changes, unanticipated signal interference, or moving obstacles can introduce real-time disturbances that are not captured in static mission planning. These factors pose significant risks to mission success and system integrity, especially when operating under strict energy, time, and safety constraints.

To address such challenges, researchers have proposed various strategies. For example, Pham et al. [71] introduced a robust adaptive control framework that compensates for nonlinear model uncertainties and external disturbances using Lyapunov-based switching laws, enabling UAVs to maintain stability under high-wind conditions. Similarly, Khan and Guivant proposed a Model Predictive Control (MPC)-based approach for UGVs that accounts for both time-varying dynamics and input constraints during terrain navigation [72]. These methods allow systems to react to real-time disturbances while minimizing performance degradation.

In the multi-agent domain, decentralized control strategies leveraging learning-based approaches such as reinforcement learning and federated learning have been explored to enhance adaptability under partial observability and limited communication bandwidth [73]. These frameworks allow each agent to learn and optimize policies based on local observations, improving overall system resilience. Moreover, fault-tolerant methods, such as those proposed by Ren and Sun, integrate fuzzy logic and neural networks to reconfigure control laws on-the-fly when sensor or actuator faults are detected [74].

Despite these advancements, gaps remain in generalizing these solutions to heterogeneous UGV-UAV teams, particularly when high levels of autonomy and cooperation are required in unstructured environments. As such, ongoing research should focus on hybrid approaches that combine robust control, adaptive learning, and multi-sensor fusion to enable safe, responsive, and efficient decision-making under real-time uncertainty.

3 Novel Solutions for UGV-UAV Integration

3.1 Unified Robotic Protocol (URP)

The Unified Robotic Protocol (URP) is proposed as an innovative hybrid communication framework designed to address the inherent communication challenges between UGVs and UAVs. URP refers to a conceptual or developed framework aimed at standardizing communication, operation, and interaction between various robotic systems [75]. This protocol serves as a universal language or set of rules enabling robots from different manufacturers, platforms, or configurations to work together seamlessly.

The URP aims to bridge these gaps by combining the strengths of ROS and MAV Link into a unified communication structure that supports seamless data transmission, synchronization of tasks, and reliable communication under variable conditions. By providing a shared interface, the URP would allow UGVs and UAVs to exchange positional data, sensor readings, and task commands with minimal latency, overcoming the limitations of existing systems that often result in delays or communication breakdowns during critical missions. Studies on helicopter dynamic models controlled via FOPID-based optimization algorithms highlight the importance of precise control and optimization in aerial robotics [76], which supports the integration aims of the URP.

Furthermore, the URP incorporates advanced security features, including blockchain technology, to ensure secure data exchange [77, 78]. Blockchain's decentralized nature guarantees that all communication is tamper-proof and transparent, offering a significant advantage when operating in high-security environments or mission-critical scenarios. The proposed protocol would improve not only the efficiency of communication but also the reliability and integrity of the entire system. Figure 5 shows the basic architecture of URP and Table 4 shows the comparison of URP with traditional communication frameworks [16, 75, 79].

In conclusion, the Unified Robotic Protocol (URP) represents a significant advancement over traditional communication frameworks like ROS and MAV Link. URP excels in providing faster communication with optimized low-latency data exchange, which is crucial for real-time coordination between UGVs and UAVs. It also ensures greater reliability by incorporating redundancy and failover mechanisms, making it more dependable in dynamic or challenging environments. Furthermore, URP's integration of blockchain technology offers enhanced security by safeguarding data integrity and preventing unauthorized access, which is a notable improvement over the more basic security features of ROS and MAV Link. While ROS and MAV Link are effective in their specific domains, URP's integrated approach to speed, reliability, and security makes it a superior choice for complex, multi-platform operations involving both UGVs

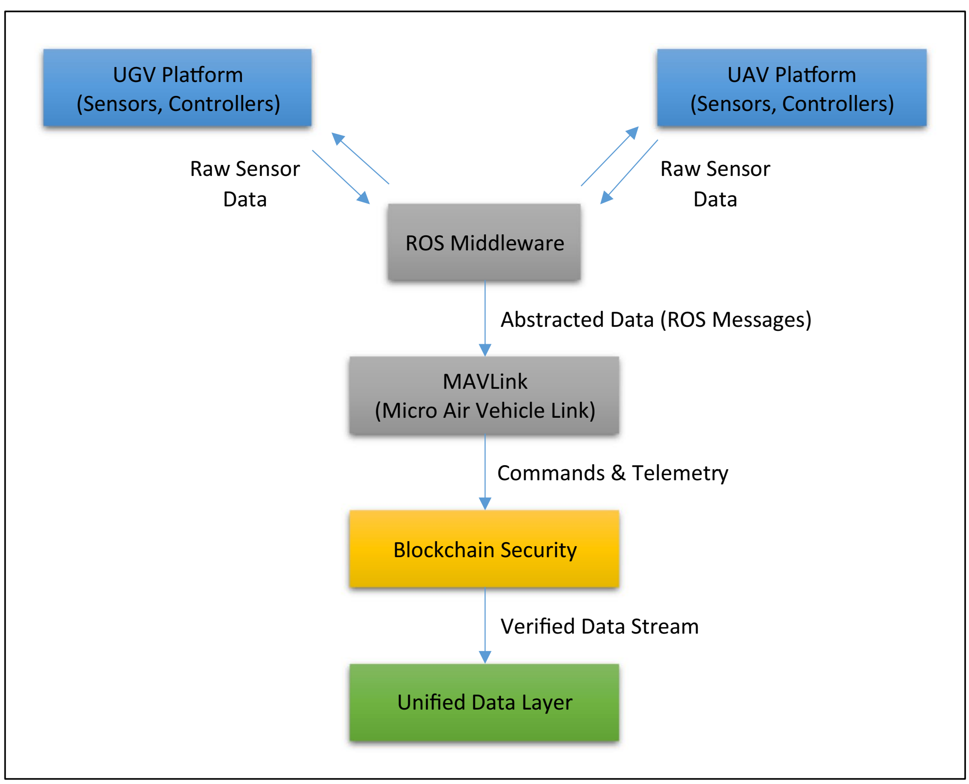


Fig. 5 Architecture of the Unified Robotic Protocol (URP), showing the interaction between UGVs, UAVs, and blockchain technology

Table 4 Comparison of URP with traditional communication frameworks (ROS vs MAV Link), highlighting advantages in speed, security, and reliability

Factor	Unified robotic protocol (URP)	ROS (Robot operating system)	MAV link protocol
Speed	- Optimized for low-latency communication between UGVs and UAVs - Incorporates multi-threaded and asynchronous data handling for faster response times	- Can experience latency due to overhead from data abstraction - Not designed for high-speed, real-time control between UGVs and UAVs	- MAV Link is lightweight and optimized for real-time communication, but primarily designed for UAVs, which may limit speed when integrated with UGVs
Reliability	- Uses robust data synchronization across both platforms - Handles communication in challenging environments with redundancy mechanisms and failover protocols to ensure consistent data exchange	- Reliability may vary with data-heavy applications - ROS often depends on the underlying network setup and lacks native failover or redundancy features in its core	- Reliable for UAV-specific operations but can struggle with complex multi-platform coordination (e.g., between UGV and UAV) - Limited fault tolerance
Security	- Integrated blockchain technology ensures data integrity, verification, and secure command transmission - Protects against tampering and unauthorized access, improving trust in communication	- ROS lacks native security features, relying on external tools and middleware for encryption and authentication - Vulnerable to network-based attacks in its base form	- MAV Link provides basic encryption (TLS/SSL) for secure communication - Primarily used for UAVs and lacks robust security mechanisms when used across multiple platforms
Interoperability	- Built to seamlessly integrate both UGV and UAV platforms through a unified framework - Supports multi-sensor fusion, enabling smooth communication across several types of robots	- Primarily designed for robot-to-robot communication on the ground or within a specific system - Interoperability between UGV and UAV may require complex setup and additional tools	- Primarily designed for UAVs, MAV Link does not inherently support seamless integration between UGVs and UAVs - Requires customization for cross-platform communication
Scalability	- Scalable across a wide range of robotic platforms and operational environments - Easily extends to accommodate additional robots (UGVs, UAVs, etc.) and sensors with minimal reconfiguration	- Scalability is possible but often requires significant custom development - Not inherently designed for cross-platform scalability (UGVs and UAVs)	- Designed for a single platform (typically UAVs) and may require significant modification to support many UGVs or other robotic platforms
Flexibility	- Highly adaptable to different mission types, from search-and-rescue to surveillance - Easily modified to integrate new sensors, control systems, and mission parameters	- Flexible for various robotic applications but requires additional integration layers for diverse types of robots - Primarily focused on ground-based robots	- Best suited for UAV operations - Less flexible for complex multi-robot coordination in diverse environments (UGV, UAV, or mixed)
Resource Efficiency	- Optimized for efficient resource usage across both UGVs and UAVs, balancing computation, communication, and energy consumption	- Can be resource-intensive, especially in large-scale systems with multiple robots - Requires more resources for high-level communication between UGV and UAV	- Low resource overhead due to lightweight design but can be limiting for complex systems requiring high-bandwidth communication or multi-robot coordination

and UAVs. Its ability to seamlessly integrate and optimize communication across these platforms underscores its potential as a unified framework for next generation.

The primary advantage of the proposed URP framework lies in its ability to unify heterogeneous communication systems under a single protocol, enabling seamless data exchange and coordinated task execution across UGVs and UAVs. Unlike existing solutions that require complex middleware or manual synchronization, URP simplifies integration through native support for both ROS and MAVLink protocols. Its blockchain-based security mechanisms ensure tamper-proof data transfer, a critical feature in high-stakes missions such as military operations and disaster response. Moreover, URP's modular architecture allows scalability and adaptability to evolving operational needs, making it a future-proof solution for multi-agent autonomous systems. These advantages significantly reduce communication latency, improve coordination accuracy, and enhance mission reliability, which are essential benchmarks in real-world deployment scenarios.

3.2 Modular Payload Systems

The development of modular payload systems represents a crucial advancement in enhancing the flexibility and adaptability of UGVs and UAVs in integrated operations. Modular payloads are designed to be interchangeable, allowing for quick reconfiguration depending on the mission requirements [12]. This dynamic adaptability ensures that the platforms can be efficiently used across a wide variety of applications, from military operations to disaster response and environmental monitoring. For example, a UAV might be equipped with a high-resolution camera and thermal sensors for reconnaissance missions, but these payloads could easily be swapped out for a LiDAR system or a communication relay system for a different task. Similarly, UGVs, which are typically used for ground-based interventions like debris removal or supply delivery, could swap out payloads such as robotic arms for more specialized tools depending on the operational need. This flexibility enhances the operational efficiency of integrated UGV-UAV systems by reducing the need for specialized platforms for each task, lowering operational costs, and increasing mission success rates. It also enables rapid deployment in dynamic, fast-changing scenarios, as the platforms can be reconfigured on the fly to respond to emergent situations.

Modular payload systems enhance mission flexibility by enabling UGVs and UAVs to be rapidly reconfigured for a broad range of operational tasks without needing specialized hardware for each mission type. This adaptability is crucial in environments where mission parameters can change unpredictably, such as in search and rescue, military surveillance, or environmental monitoring. For instance, in a disaster response operation, a UAV can switch between a thermal imaging payload for locating survivors and a communication relay payload to support real-time coordination among ground teams. Similarly, a UGV can alternate between robotic manipulators for debris removal and sensor suites for structural integrity assessments. These modular systems often use standardized mechanical and electrical interfaces, allowing field operators to swap payloads with minimal tools and technical expertise. The result is a flexible, scalable platform that maximizes mission effectiveness while minimizing logistical burden and downtime. Moreover, modular payload architectures support "plug-and-play" compatibility, where payloads can self-identify and configure upon connection. This feature, combined with software-defined control systems, ensures that UGVs and UAVs can adapt not only physically but also functionally, based on mission goals. As operations grow in complexity, such flexibility becomes critical in maintaining operational tempo and responsiveness.

The significance of modular payload systems lies in their ability to transform UAVs and UGVs into highly versatile and efficient tools for a wide range of applications [12, 80]. These systems enhance versatility by enabling a single platform to perform multiple roles, eliminating the need for specialized vehicles tailored to specific tasks. This adaptability proves invaluable in dynamic operational environments, where mission requirements often shift rapidly.

Moreover, modular payloads contribute to cost efficiency [81]. By using a single reconfigurable platform instead of multiple dedicated systems, organizations can reduce both procurement and maintenance expenses. Rapid deployment is another critical advantage, as these systems allow operators to reconfigure payloads in the field with minimal downtime, ensuring continuous mission readiness.

Future-proofing is another key benefit of modular designs [82]. As technology evolves, platforms equipped with interchangeable payloads can seamlessly integrate new capabilities without requiring a complete system replacement. This scalability ensures the longevity of the technology while keeping it relevant for emerging needs.

Finally, modular payload systems offer a significant strategic advantage. Their ability to adapt quickly to different scenarios makes them highly valuable across industries such as defence, emergency response, and logistics, where flexibility and rapid adaptation are essential [83, 84]. This novel approach represents a departure from traditional, fixed-design systems, marking a shift toward more sustainable, multipurpose solutions in autonomous technology. Figure 6 shows the possibilities of modular load systems of UAV and UGV integration.

Traditional UAVs and UGVs were often built with fixed payloads tailored to specific tasks, limiting their utility

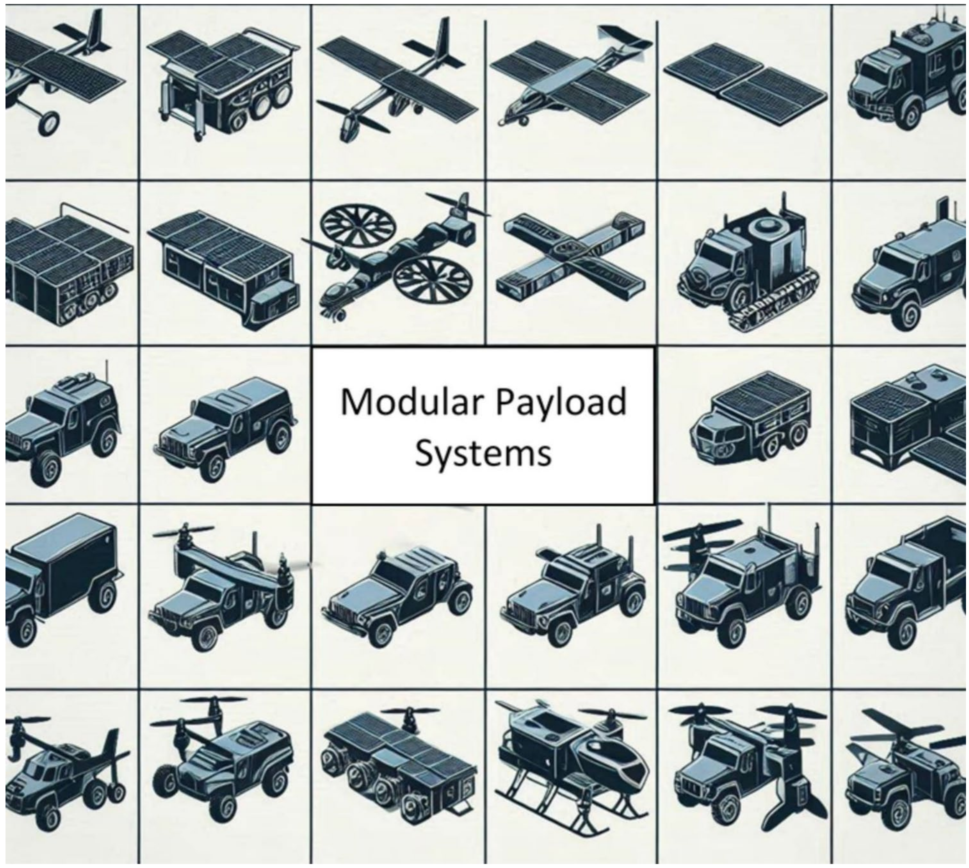


Fig. 6 Modular payload systems for UGVs and UAVs, illustrating the flexibility of each platform with interchangeable tools and sensors (Source: Generated with AI, November 29, 2024, at 9:33 AM)

across multiple scenarios. The modular approach breaks this mold, enabling a"plug-and-play"design philosophy. This novelty not only optimizes operational efficiency but also aligns with the growing demand for multipurpose, sustainable solutions in autonomous systems.

Table 5 shows the comparative performance and cost-efficiency of modular systems in UAVs and UGVs against single-purpose payloads, highlighting key operational and economic distinctions. Modular systems generally exhibit higher initial costs due to the advanced engineering required for interoperability and universal compatibility [85]. However, their inherent flexibility allows them to adapt to diverse mission profiles through rapid payload reconfiguration, which single-purpose payload systems cannot achieve without deploying entirely separate units [86]. This adaptability provides modular systems with a strategic advantage in dynamic operational environments [2].

From a cost perspective, modular systems demonstrate superior long-term efficiency [87]. While the initial investment may be higher, shared platforms and interchangeable components reduce maintenance costs and simplify logistics, such as spare part inventories and repairs. Single-purpose systems, in contrast, often incur greater cumulative costs over time, particularly when missions necessitate multiple platforms to fulfil varied roles. Modularity also ensures a higher return on investment (ROI) as upgrades can be integrated without replacing the entire platform, thereby extending the system's lifecycle.

Operationally, modular systems excel in reducing mission downtime by enabling quick swapping of payloads, a capability particularly valuable in time-sensitive scenarios such as disaster response or multi-phase military operations [88]. Although single-purpose payloads may offer optimized performance and lighter system weight for their specific task, this comes at the cost of reduced flexibility and limited scalability. In situations requiring specialization, single-purpose systems may be advantageous, but they are less adaptable to evolving mission requirements.

In conclusion, modular systems emerge as a more practical and cost-efficient solution for scenarios that demand versatility, scalability, and long-term value. They are particularly well-suited for applications in dynamic environments where the ability to perform multiple roles is critical. Single-purpose payloads, while effective in fulfilling narrowly defined tasks, lack the adaptability and economic sustainability of modular platforms in diverse and unpredictable mission profiles. Consequently, modular systems

Table 5 Comparison of modular payload systems versus single-use payloads, emphasizing cost-efficiency, operational flexibility, and mission success rates

Criteria	Modular systems (UAV & UGV)	Single-purpose payloads
Initial Cost	Higher due to the need for modular hardware and interfaces	Lower since components are tailored for a specific purpose
Operational Flexibility	Very high: allows adaptation to diverse missions by swapping payloads	Limited: constrained to the original design purpose
Mission Variety	Can perform multiple roles (e.g., surveillance, delivery, reconnaissance)	Limited to one role per unit, requiring multiple platforms for varied tasks
Payload Integration	Plug-and-play modularity simplifies swapping; standards ease integration	Fixed payloads require customization or rebuilding for changes
Maintenance Costs	Lower long-term: shared parts simplify repairs and spare stock	Higher due to unique parts and design-specific spares
Deployment Speed	High: modular designs enable rapid reconfiguration on-field	Slower: may require deploying separate systems for different tasks
System Weight	Typically, heavier due to modular connectors and universal compatibility requirements	Lighter, as designed for a single-use case
Power Efficiency	Slightly lower since modular systems may carry unused capacities	Higher for a given task due to optimized payload design
Upgradability	Excellent: modularity allows upgrades without replacing the entire system	Poor: upgrades often require full replacement or redesign
Scalability	High: modularity supports fleet expansion with standardized parts	Limited scalability due to bespoke designs
Mission Downtime	Minimal: quick swapping of payloads reduces the time between missions	Higher: requires separate vehicles or manual reconfiguration downtime
Cost per Mission Type	Lower: shared platforms reduce costs when used for diverse tasks	Higher: new systems are needed for each unique mission
Long-Term ROI	High: cost amortized over a wide range of missions and longer lifecycle	Lower: frequent replacements or additional purchases increase expenses
Examples	UAVs: DJI Matrice series; UGVs: Milrem THeMIS	UAVs: DJI Phantom; UGVs: fixed-purpose bomb disposal robots

provide a strategic advantage in maximizing operational efficiency and resource allocation over extended periods.

3.3 AI and Swarm Intelligence for Task Allocation

Artificial intelligence (AI) and swarm intelligence are key innovations in optimizing task allocation and coordination within integrated UGV-UAV systems. These technologies enable autonomous decision-making, allowing UGVs and UAVs to work collaboratively with minimal human intervention. In disaster zones, UAVs use onboard AI to identify survivors, then communicate with UGVs to deliver supplies or clear paths—without human coordination. This real-time dynamic tasking is enabled through swarm behavior models and peer-to-peer coordination [30, 89].

In addition to AI, swarm intelligence plays a significant role in facilitating decentralized coordination [90]. Inspired by biological systems like ant colonies or bird flocks, swarm intelligence algorithms allow multiple UGVs and UAVs to collaborate without a central controller, optimizing their collective behaviour based on simple local rules. This decentralized approach to task allocation reduces the risk of bottlenecks and improves the scalability of the system, making it possible to deploy large numbers of unmanned vehicles in complex environments, such as search-and-rescue operations in disaster zones or large-scale surveillance tasks.

Figure 7 provides a schematic representation of the role of AI and swarm intelligence in enabling autonomous task allocation and coordination among UAVs and UGVs. The layout illustrates a decentralized system where individual agents—both aerial and ground-based—collaborate dynamically to achieve mission objectives. In the upper section of the diagram, UAVs are depicted as aerial agents scouting the environment, identifying tasks such as mapping terrain, detecting targets, or marking areas of interest. These tasks are represented in the middle layer, symbolizing the transitional phase where information gathered by UAVs is processed and relayed to other agents within the system. The lower section shows UGVs as ground-based agents responsible for executing tasks that require direct physical interaction with the environment, such as delivering supplies, navigating rough terrain, or collecting samples.

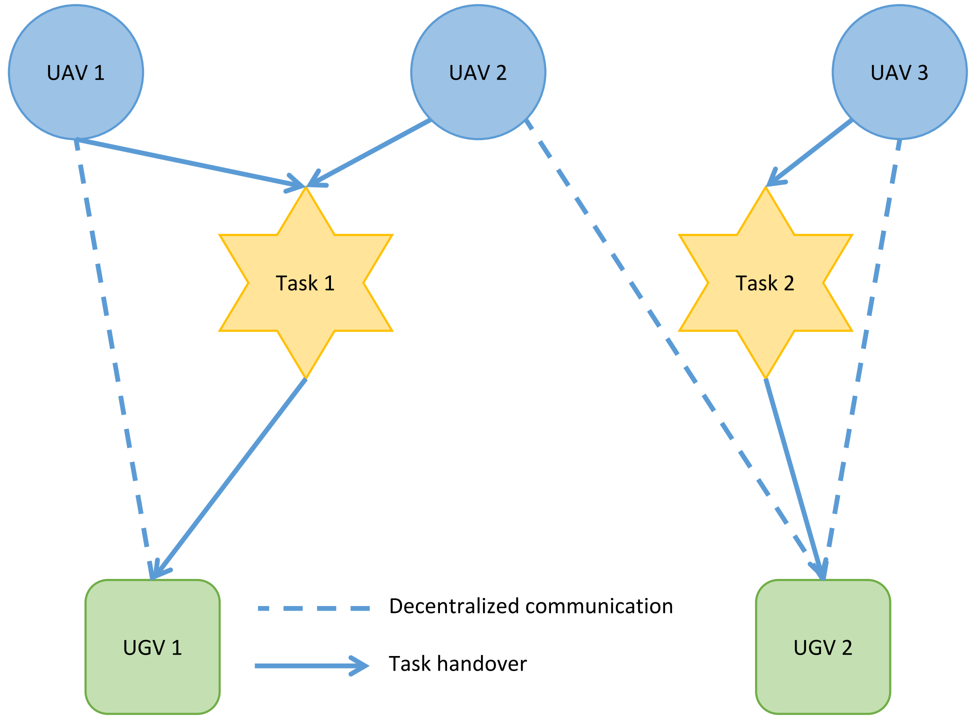


Fig. 7 Swarm intelligence task allocation in UGV-UAV systems illustrates the decentralized coordination between multiple agents

AI algorithms embedded within each UAV and UGV facilitate the decision-making process required for effective task allocation. Specifically, swarm intelligence principles, such as local communication and emergent behaviour, allow agents to operate without a central controller. This is depicted in Fig. 7 by dashed lines that represent the decentralized peer-to-peer communication network, which ensures that agents share critical information, such as task status or environmental changes, in real-time. Solid arrows in the diagram highlight the dynamic nature of task allocation, demonstrating how UAVs pass actionable insights, such as identified tasks, to UGVs, which then conduct ground-level operations.

The autonomy and adaptability of this system are core benefits of swarm intelligence. Each agent perceives its local environment, evaluates available tasks, and communicates with others to allocate roles based on proximity, capability, or priority. The absence of a single point of control enhances the system's robustness, allowing it to adapt to changing conditions or the failure of individual units without compromising the mission. For instance, if a UAV identifies a new area of interest, the system autonomously reallocates tasks to ensure coverage, while idle UGVs are assigned new roles to maintain operational efficiency. Hybrid optimization approaches, such as the integration of FOPID-TID controllers with advanced metaheuristic algorithms like HAOAROA, have demonstrated significant improvements in UAV path planning efficiency and stability in uncertain environments [91]. Other than that, hybrid intelligent control strategies like the FOPDD + I controller combined with ANFIS models have shown promising results in voltage regulation scenarios, offering parallels to AI-driven control in UAV-UGV systems [92]. This decentralized coordination, powered by AI and swarm intelligence algorithms, enables seamless collaboration between UAVs and UGVs in complex and dynamic environments. The approach is particularly advantageous in scenarios such as disaster response, where rapid task allocation, adaptability, and resilience are critical to mission success.

Recent developments in model-free and uncertain artificial intelligence-based control laws and path planning techniques have shown promising results in enhancing UAV autonomy, especially in fault-tolerant and constrained scenarios. Notably, bioinspired machine learning algorithms have been effectively utilized for minimum distance and minimum time optimal path planning, even under faulty UAV conditions, ensuring safe and efficient trajectory generation [93]. In parallel, the introduction of robust reduced-order Thau observers coupled with adaptive fault estimators provides a significant advancement in UAV fault diagnosis and resilience, enabling reliable operation in uncertain environments [94]. These approaches not only complement existing swarm intelligence-based strategies but also pave the way for hybrid control frameworks that integrate optimal path planning with decentralized AI-based coordination. Future implementations of UGV-UAV teams could greatly benefit from incorporating such fault-resilient control systems to enhance reliability and operational safety in real-time applications.

3.4 Integration of Existing Systems

To ensure seamless operation between UGVs and UAVs, the integration of existing architectures is critical. This includes not only the synchronization of hardware components but also the harmonization of software frameworks and data protocols. Systems such as the Android Team Awareness Kit (ATAK), widely utilized for real-time situational awareness and mission coordination, offer a robust platform for integrating disparate architectures. ATAK can serve as a common interface, enabling UGVs and UAVs to share operational data, such as positional information, sensor outputs, and mission parameters, in real-time [95].

Translation modules could further enhance this integration by acting as intermediaries between different protocols, such as the Robot Operating System (ROS) for UGVs and MAV Link for UAVs. These modules would encapsulate and translate commands and telemetry data into a unified format, ensuring compatibility while maintaining low latency. Figure 8 below illustrates the role of a translation module in facilitating communication between UGV and UAV systems through a shared platform like ATAK.

Integrating existing systems through platforms like ATAK and translation modules provides several advantages. First, it enhances interoperability by reducing the complexities associated with protocol incompatibilities [96]. ATAK serves as a standardized interface, allowing UGVs and UAVs to communicate seamlessly, ensuring that disparate systems work together efficiently. This integration improves operational efficiency by minimizing communication delays, which is particularly crucial in dynamic environments such as military operations or disaster response scenarios. Additionally, the use of translation modules ensures that data is processed and transmitted with minimal latency, allowing for real-time coordination and decision-making.

Another significant benefit of this integration is its scalability and adaptability [95]. The use of modular systems and common platforms like ATAK enables the easy incorporation of additional robotic platforms or sensor systems without requiring substantial changes to the underlying architecture. As the demands of missions evolve, the system can rapidly adjust by adding new sensors or vehicles, thus maintaining operational flexibility. Moreover, shared interfaces between the systems improve situational awareness by consolidating all relevant information in a central dashboard, allowing operators to monitor and control multiple platforms simultaneously, reducing cognitive load and enhancing decision-making accuracy.

While the integration of existing systems offers significant advantages, it also presents several challenges. One of the primary obstacles is ensuring data synchronization across different platforms [97]. For UGVs and UAVs to operate effectively together, they must share real-time data with accurate time stamps, which requires robust synchronization mechanisms. Without this synchronization, discrepancies in task execution or situational awareness could arise, potentially compromising mission success.

Another challenge is the efficient management of data flow, especially when integrating high-bandwidth systems like LiDAR or high-resolution video streams [98]. As the volume of data transmitted between systems increases, ensuring that the communication infrastructure can manage the load without introducing latency becomes critical. Additionally, security remains a significant concern. Platforms like ATAK, which serve as central hubs for coordination, must be protected from cyber threats. Incorporating blockchain technology could address these concerns, providing secure, tamper-proof communication channels between systems, ensuring data integrity, and preventing

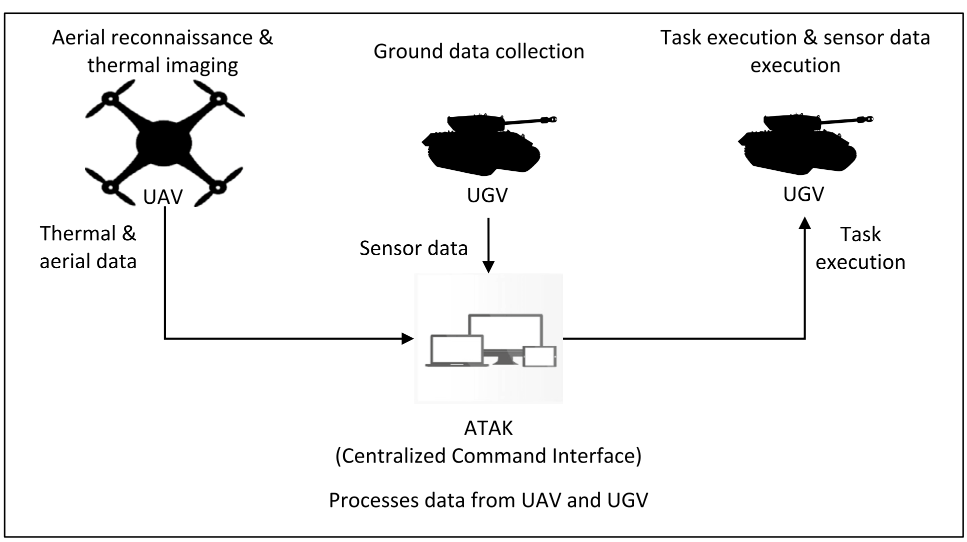


Fig. 8 UGV-UAV integration using ATAK

unauthorized access. Overcoming these challenges will be essential to realizing the full potential of integrated UGV-UAV operations.

4 Real-world Applications and Case Studies

The integration of UGVs and UAVs in real-world applications has demonstrated substantial benefits, particularly in military, disaster response, and surveillance operations. Case studies highlight how the complementary strengths of these platforms can improve mission outcomes, reduce human risk, and enhance operational efficiency. However, challenges related to communication, sensor interoperability, and environmental constraints continue to impede the seamless integration of these systems. As research in communication protocols, sensor fusion techniques, and autonomous decision-making continues to progress, the potential for fully integrated UGV-UAV systems to revolutionize various domains becomes increasingly tangible. The integration of advanced technologies such as AI, blockchain, and quantum computing could further optimize UGV-UAV collaboration, enabling these systems to operate with greater autonomy, resilience, and effectiveness in complex environments.

4.1 Military Operations

The integration of UGVs and UAVs in military operations offers significant advantages in terms of enhanced situational awareness, reduced risk to human life, and improved operational efficiency. Military applications often require real-time intelligence, rapid mobility, and precise execution of tasks. UAVs equipped with high-resolution cameras and thermal sensors can provide persistent aerial surveillance, allowing commanders to monitor large areas, identify potential threats, and track enemy movements. In contrast, UGVs can perform tasks that are challenging for UAVs, such as navigating through difficult terrain or carrying heavy payloads like ammunition, supplies, and equipment.

A notable case study illustrating the effective integration of UGVs and UAVs occurred during urban warfare exercises. UAVs were deployed to provide aerial reconnaissance and real-time video feeds to ground forces [99], while UGVs were used to deliver supplies and neutralize potential threats in hazardous zones [100] (Fig. 9). The integration of these platforms significantly reduced the time required to identify threats and deliver critical supplies, while minimizing human exposure to dangerous situations. However, challenges such as communication latency and the need for synchronized task execution were identified. To address these, improvements in the communication protocols, such as the proposed Unified Robotic Protocol (URP), could enhance

Fig. 9 Coordination between UAVs providing reconnaissance and UGVs delivering supplies during an urban warfare exercise (Source: Generated with AI, November 29, 2024, at 11:36 AM)

the coordination between platforms, ensuring faster and more efficient decision-making.

4.2 Disaster Response

UGV-UAV integration has proven to be highly effective in disaster response scenarios, where rapid situational awareness and efficient deployment of resources are critical. During natural disasters such as earthquakes, floods, or wildfires, UAVs can quickly survey large areas, providing real-time aerial imagery of affected regions. This capability allows emergency responders to identify hazards, locate survivors, and assess damage from a safe distance. UGVs, on the other hand, can navigate the affected terrain and perform critical tasks on the ground, such as debris removal, delivering medical supplies, or conducting search-and-rescue operations in areas inaccessible to humans [101].

In the aftermath of the Emilia Romagna earthquake, a UAV provided exterior visual information of a church tower for structural damage assessment. Simultaneously, a UGV was used to explore the interior of a dome to assess the state of important religious artifacts. This dual deployment enabled a comprehensive evaluation of both external and internal structural integrity, aiding in prioritizing rescue and preservation efforts [101]. This integration significantly improved response times, as emergency crews could access real-time data from the UAVs and deploy UGVs to specific locations with greater precision.

Despite the success, challenges related to battery life and environmental constraints, such as unstable ground and limited visibility, hindered prolonged operation. Future advancements in battery technologies, such as solid-state batteries or solar-powered UAVs, could further extend the operational time of these systems, enhancing their effectiveness in disaster response. Table 6 shows the comparison of UAV and UGV roles in disaster response [2].

UAVs are particularly effective in disaster response scenarios that require rapid coverage of large areas, such as aerial surveying, reconnaissance, and quick supply delivery. Their ability to navigate over obstacles and access difficult-to-reach areas makes them invaluable for providing real-time situational awareness. However, their limited payload capacity and vulnerability to adverse weather conditions can constrain their usefulness in certain tasks. Additionally, UAVs cannot interact physically with their environment, which limits their ability to assist directly in tasks like debris removal or survivor extraction [102].

UGVs, by contrast, are essential for tasks that require direct interaction with the environment. They can navigate through difficult terrain, move heavy objects, and conduct precise tasks such as debris removal and survivor rescue. UGVs are also better suited for carrying heavier payloads, such as food, water, or medical supplies. However, they are slower than UAVs and face significant mobility challenges in areas with dense rubble or complex obstacles. UGVs also require real-time data from UAVs to effectively assess the terrain and navigate complex environments, making them reliant on aerial platforms for reconnaissance and environmental assessment [101].

In conclusion, the strengths, and limitations of both UAVs and UGVs highlight the complementary nature of their roles in disaster response. UAVs are ideal for rapid exploration, aerial surveying, and high-level situational awareness, while UGVs provide the physical presence needed for ground-level interventions such as debris clearing and supply delivery. Together, these platforms form a synergistic team, enhancing the overall effectiveness of disaster response efforts by combining speed, flexibility, and physical capacity in a comprehensive and adaptive approach.

4.3 Surveillance and Monitoring

The combined use of UGVs and UAVs in surveillance and monitoring applications provides a comprehensive solution for tracking large-scale activities in both urban and rural environments. UAVs offer broad coverage capabilities, ideal for monitoring vast areas such as borders, wildlife reserves, or critical infrastructure [13]. These platforms can be equipped with a variety of sensors, including infrared cameras, LiDAR, and environmental sensors, to provide high-resolution data for real-time decision-making. UGVs complement UAVs by performing close-range inspection tasks that require direct interaction with the environment, such as detecting contraband at border checkpoints, inspecting pipelines, or assessing the structural integrity of buildings and bridges [2].

A key example of successful UGV-UAV integration in surveillance occurred during border patrol operations, where UAVs provided aerial surveillance of large border areas, detecting potential illegal crossings or contraband [103]. Once a potential threat was identified, UGVs were deployed to specific locations for detailed inspection and intervention. The integration of these platforms allowed for a more efficient allocation of resources, as UAVs could cover a broad area while UGVs oversaw more localized tasks. However, challenges such as signal interference, data synchronization, and the need for seamless coordination between the two platforms remain. To address these issues, the implementation of advanced communication frameworks, like the Unified Robotic Protocol, and the development of hybrid control systems could enhance coordination, ensuring better synchronization between UAVs and UGVs in real-time. Figure 10 shows the flowchart of the surveillance process for the integration of UAV and UGV.

Table 6 A comparison of UAV and UGV roles in disaster response

Scenario	UAV			UGV		
	Role	Advantages	Limitations	Role	Advantages	Limitations
Aerial Surveying	UAVs provide rapid mapping and aerial imagery of affected areas	Cover large areas quickly, unaffected by ground obstacles	Limited battery life; affected by adverse weather conditions	UGVs are not suitable for aerial surveying	Not applicable	Cannot perform aerial tasks; dependent on UAV data for terrain analysis
Debris Removal	UAVs have minimal to no role in debris removal	Not applicable	Cannot engage with physical obstacles; limited payload capacity	UGVs equipped with manipulators can clear debris and move heavy objects	Can operate in harsh conditions and physically manipulate the environment	Limited mobility in extremely uneven terrain; slower than UAVs for wide coverage
Survivor Rescue	UAVs identify survivors through thermal imaging and relay locations	Quick identification from a safe altitude; unaffected by terrain	Limited interaction with survivors; may struggle with dense obstructions	UGVs can transport supplies to survivors or extract them from dangerous areas	Can provide physical assistance, such as carrying survivors or supplies	Limited operational speed; may encounter difficulty in complex terrain
Search and Recon	UAVs scout for safe paths, hazards, and accessible routes	Speedy exploration of large areas; can bypass obstacles like water or cliffs	May miss ground-level details or access constrained spaces	UGVs verify UAV reconnaissance on the ground and confirm hazards	Ground verification adds accuracy and adaptability to dynamic changes	Slower reconnaissance; restricted by physical terrain challenges
Supply Delivery	UAVs deliver small, lightweight packages like medicine or communication gear	Faster delivery over challenging terrain; bypasses road blockages	Limited by payload capacity and weather conditions	UGVs deliver heavier supplies such as food, water, or medical equipment	Higher payload capacity; safer delivery in dangerous environments	Slower than UAVs; limited to terrain accessibility
Environmental Monitoring	UAVs monitor air quality, gas leaks, or radiation levels from above	Quick, high-level monitoring without physical exposure to danger	Limited to aerial measurements; less accurate for ground-level conditions	UGVs perform ground-level environmental sampling and inspections	Precise, ground-level data collection in hazardous environments	Limited sensor range; restricted by terrain in hazardous areas

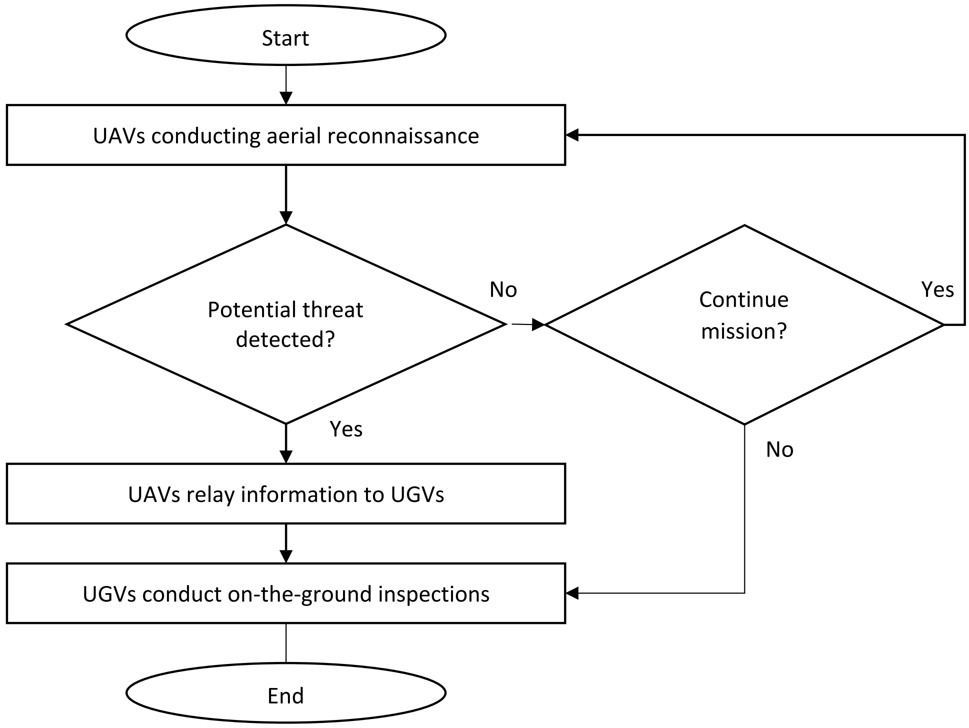


Fig. 10 Surveillance process: UAV and UGV Coordination

5 Emerging Technologies and Future Directions

5.1 Blockchain for Secure Communication

The integration of UGVs and UAVs relies heavily on secure communication to ensure real-time coordination and data integrity. Traditional communication protocols, such as Robot Operating System (ROS) for UGVs and MAV Link for UAVs, face limitations in ensuring both security and reliability, particularly in high-stakes environments like military operations or disaster response. Blockchain technology presents a promising solution to these issues, offering a decentralized framework for data exchange that ensures transparency, immutability, and security.

In the context of UAV-UGV communication, blockchain offers several advantages. UAVs and UGVs can securely transmit mission data, sensor readings, and instructions through blockchain nodes, which validate and store the information. Blockchain's role in UGV-UAV integration is pivotal for preventing data tampering or unauthorized access during mission-critical operations [104]. Each communication transaction between UGVs and UAVs can be recorded as a block in a distributed ledger, ensuring that all transmitted data is traceable and auditable. This approach not only secures communication channels but also establishes trust between unmanned systems operating in potentially hostile or compromised environments.

Another important aspect of blockchain is its transparency and auditability [105]. Every transaction or data exchange is recorded in an immutable ledger, providing a verifiable trail of communication. In operations requiring transparency—such as military missions or disaster recovery efforts—this auditability ensures that the system's actions are traceable and dependable. The blockchain's auditability allows mission planners or commanders to trace actions and verify that vehicles are following the correct procedures. This makes it easier to trace and audit the interactions between UAVs, UGVs, and blockchain nodes, offering a prominent level of accountability. In high-stakes scenarios such as military operations or disaster response, this transparency is critical for ensuring that the operation proceeds as planned and that all vehicles are performing their tasks correctly. The system's secure, tamper-proof data storage also helps maintain accountability throughout the mission.

Another one of the most important features of blockchain is decentralization [106]. Unlike traditional communication systems, where a central authority manages the flow of information, blockchain enables a distributed network of nodes (computers or devices) to share the responsibility for validating and storing data. In UAV-UGV operations, this decentralized nature ensures there is no single point of failure, meaning that if one node goes down or is compromised, the rest of the system remains operational. This guarantees continuous and secure communication between the vehicles without dependence on a central server. Blockchain's decentralized nature ensures that there is no crucial point of failure, and the data is protected through encryption and validation mechanisms. This secure communication allows the UAVs and UGVs to operate autonomously and efficiently, without the risk of data tampering or unauthorized access. Furthermore, blockchain enables both vehicles to engage in mission coordination, where UAVs provide real-time data to blockchain nodes, and UGVs access that validated data to execute their tasks. For example, a UAV may send sensor data indicating an obstacle or hazard, and the UGV can retrieve that information from the blockchain to take appropriate action. Blockchain's ability to store and validate this data in a transparent and immutable manner ensures that all actions are based on accurate and reliable information.

In addition to ensuring security, blockchain can also help in creating a multi-agent coordination system, where multiple UGVs and UAVs collaborate under a unified framework [106]. The transparency of blockchain ensures that all participating agents have access to the same data and can make decisions based on real-time information, reducing the likelihood of communication breakdowns. The development of permissioned blockchains could further enhance security by allowing only authorized entities to access the network, ensuring that only trusted unmanned vehicles communicate with one another. Blockchain relies on consensus mechanisms, such as Proof of Work (PoW), Proof of Stake (PoS), or Practical Byzantine Fault Tolerance (PBFT), to validate transactions [107]. These mechanisms require multiple nodes in the network to reach an agreement about the validity of a transaction before it is recorded in the blockchain. In UAV-UGV communication, this ensures that data exchanged between vehicles and blockchain nodes is verified by the network, making it impossible for fraudulent or incorrect data to be accepted into the system. Figure 11 shows the basic framework of blockchain-based communication of UAV-UGV.

Blockchain also provides immutability, which means that once data is recorded on the blockchain, it cannot be altered or deleted [108]. This feature is achieved through cryptographic hashing, which links each block of data to the previous one. In the case of UAVs and UGVs, this ensures that mission data, sensor readings, or instructions sent between vehicles cannot be tampered with. The data is stored securely and remains authentic, providing an elevated level of confidence in the exchanged information. The use of cryptographic security in blockchain technology further protects communication and provides authentication and authorization capabilities [109]. Every piece of data exchanged between the UAVs, UGVs, and blockchain nodes is encrypted before it is sent across the network. This encryption protects sensitive information—such as mission parameters or sensor data—from being intercepted by unauthorized parties. Additionally, the blockchain employs public–private key pairs to ensure that only authorized systems can access or decrypt the information, preventing unauthorized access to critical data. The comparison between traditional communication protocols and blockchain-enabled communication in Table 7 highlights key differences in terms of security, scalability, and latency.

In-depth reviews of blockchain data architectures reveal ongoing challenges and innovations that directly align with the proposed use of blockchain for UAV-UGV data integrity and operational transparency [110]. Traditional communication protocols offer advantages in terms of lower latency and easier scalability for smaller systems, but they come with inherent security risks due to their reliance on centralized infrastructure. Blockchain-enabled communication, while introducing some latency because of its consensus mechanisms, offers substantial benefits in terms of security and scalability. By decentralizing control and ensuring that data cannot be tampered with, blockchain provides a more secure and resilient system. These attributes make blockchain particularly well-suited for complex, security-sensitive environments, such as those involving UAVs and UGVs, where secure, reliable communication is critical.

In conclusion, blockchain technology offers significant benefits for secure communication between UAVs, UGVs, and other systems. By leveraging features like

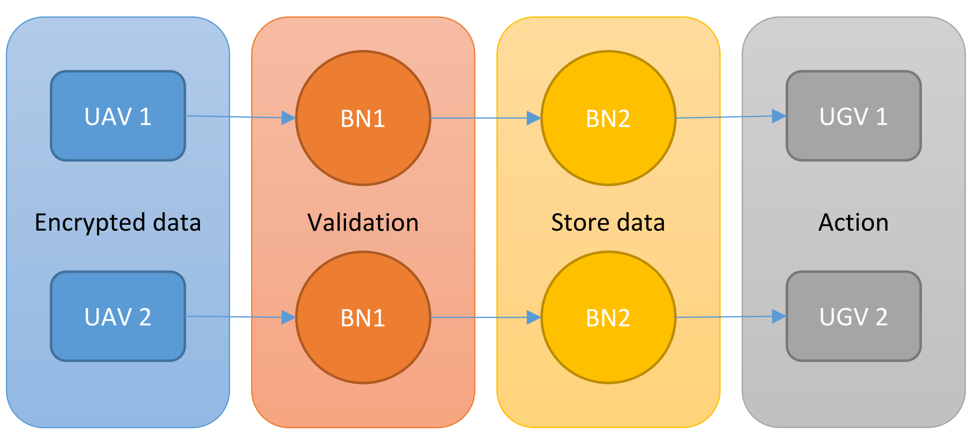


Fig. 11 A blockchain-based communication framework between multiple UGVs and UAVs shows secure data transmission across nodes

Table 7 Comparison of traditional communication protocols with blockchain-enabled communication in terms of security, scalability, and latency

Factor	Traditional communication protocols	Blockchain-enabled communication
Security	Security depends on centralized servers, making the system vulnerable to hacking and data breaches. Authentication and data integrity are managed by a central authority, which can be compromised	Blockchain ensures security through decentralization, cryptographic encryption, and immutability. Data is stored across distributed nodes, making it tamper-resistant and resilient to attacks
Scalability	Scalability can be a limitation in traditional systems, as adding more nodes or devices often requires upgrading centralized infrastructure and can lead to bottlenecks	Blockchain is highly scalable in decentralized systems. As more nodes are added, the network can maintain performance, and scalability can be achieved through various consensus mechanisms
Latency	Traditional communication protocols can suffer from higher latency due to centralized servers and network congestion. Transmission delays may occur depending on the number of intermediary devices or servers	Blockchain typically incurs higher latency due to the consensus mechanism and the time required for block validation and transaction confirmation. However, technologies like Layer-2 solutions or off-chain data storage are used to reduce latency

decentralization, immutability, cryptographic security, consensus mechanisms, transparency, and smart contracts, blockchain enables secure, reliable, and transparent operations in complex environments. As UAVs and UGVs become more integrated into industries such as military, surveillance, and disaster recovery, blockchain will continue to play a key role in ensuring that these autonomous systems can communicate securely and operate efficiently.

5.2 Quantum Computing for Optimization

Quantum computing holds the potential to revolutionize the coordination and optimization of UGVs and UAVs by enabling the rapid processing of complex tasks that are currently beyond the capabilities of classical computing systems [111–114]. One of the core challenges in UGV-UAV integration is the optimization of task allocation and resource management, especially when coordinating multiple autonomous platforms across large-scale missions. Quantum computing could significantly improve the efficiency of these processes, particularly in scenarios where the computational requirements are too high for conventional systems to handle [111].

Quantum computers can solve optimization problems exponentially faster than classical systems due to their ability to process multiple solutions simultaneously using quantum bits (qubits) [114]. In the context of UGV-UAV integration, this could be applied to real-time mission planning, where quantum algorithms are used to allocate tasks based on dynamic environmental conditions, platform capabilities, and mission priorities. For example, in a disaster response scenario, quantum computing could optimize the deployment of UAVs and UGVs to cover the maximum area in the shortest amount of time while balancing battery life, payload capacity, and environmental hazards.

Moreover, quantum computing can enhance sensor fusion by providing faster data processing [115], allowing UGVs and UAVs to combine information from disparate sensors (e.g., thermal imaging, LiDAR, radar) more efficiently. This rapid data integration would lead to better situational awareness and more accurate decision-making, especially in complex environments where quick responses are critical. Table 8 shows the comparison of classical algorithms and quantum algorithms.

In the context of processing time, classical algorithms often become slow and inefficient as the size and complexity of the coordination problem increase [116]. With problems involving many variables or agents, such as task allocation between multiple UAVs and UGVs, classical algorithms experience exponential growth in computation time. In contrast, quantum algorithms offer the promise of drastically reducing processing time by leveraging quantum parallelism. By exploring multiple viable solutions simultaneously, quantum systems can find optimal or near-optimal solutions

Table 8 Comparison of processing times and solution accuracy between quantum and classical algorithms for coordination problems

Factor	Classical algorithms	Quantum algorithms	Source
Processing Time	Slower for large, complex problems; exponential growth	Faster, uses quantum parallelism for quicker solutions	[116]
Solution Accuracy	Exact or approximate; accuracy degrades with complexity	High probability of near-optimal solutions	[116]
Scalability	Struggles with large systems; requires more resources	Scalable; manages large, complex problems efficiently	[117]
Complexity Handling	Can manage moderate complexity but struggles with large-scale or uncertain problems	Better suited for high-dimensional, complex tasks	[118]
Real-time Decision Making	Slower to adapt to dynamic environments	Faster adaptation with quantum annealing or machine learning	[119]

much faster than classical methods, especially for large-scale problems.

Solution accuracy in classical algorithms depends on whether the algorithm aims for an exact or approximate solution. Exact algorithms (like dynamic programming) can guarantee optimal solutions but are often impractical for large problems, while heuristic or approximate algorithms provide good but not always optimal solutions [116]. As the complexity of the problem increases, classical heuristics may deliver less accurate solutions. Quantum algorithms, while not always guaranteeing exact solutions, have the potential to find near-optimal solutions with a high probability. The accuracy of quantum algorithms will improve as quantum hardware evolves, providing more precise solutions over time.

When examining scalability, classical algorithms encounter significant difficulties when the number of agents in the coordination task (e.g., UAVs or UGVs) increases [117]. The computational requirements for solving problems involving many interacting agents or variables become prohibitive. Quantum algorithms, due to their inherent ability to oversee parallel processing, offer a scalable solution. Quantum systems can process exponentially more combinations of solutions, making them better suited for large-scale problems. The application of game-theoretic models such as NB Theory provides new perspectives on multi-agent negotiation and decision-making, which could be extended to decentralized UAV-UGV mission planning [120].

Complexity handling is another area where quantum algorithms show potential advantages. Classical systems struggle with complex coordination tasks that involve many interacting variables, especially when these problems are dynamic or involve uncertain parameters. Quantum computing, using its ability to create superposition states and entangle information, can efficiently explore and solve these complex problems with greater precision, handling high-dimensional spaces more effectively [118]. Evaluating FOPID criteria through Lyapunov-based approaches provides a strong theoretical basis for optimizing control trade-offs in complex systems like the AFCS [121], and such foundational work could complement quantum-enhanced optimization frameworks.

Finally, when considering real-time decision-making, classical systems may be slower in adapting to changing conditions because they often require sequential computations [119]. In contrast, quantum computing's ability to perform real-time computations with minimal delays—thanks to quantum annealing or quantum machine learning—makes it ideal for scenarios that require quick and adaptive coordination between multiple agents, such as in UAV and UGV fleet management during dynamic missions.

Quantum computing offers significant improvements in processing time, solution accuracy, scalability, and handling of complex tasks. As quantum technology continues to advance, its ability to manage large, complex problems in real-time will make it an invaluable tool for optimizing the coordination and task allocation of autonomous systems in a variety of domains, from defence to disaster recovery.

5.3 Advanced Battery Technologies

Battery life remains one of the most significant limitations for both UGVs and UAVs, especially during long-duration missions in challenging environments. The energy demands of these platforms, particularly UAVs, can restrict their operational time, preventing them from fully exploiting their capabilities [122]. Advanced battery technologies are crucial for enhancing the performance and autonomy of integrated UGV-UAV systems.

Recent advancements in solid-state batteries, which replace the liquid electrolyte with a solid one, have significantly improved energy density, safety, and longevity compared to conventional lithium-ion batteries [123]. Solid-state batteries could potentially double the energy density of current batteries, allowing UAVs and UGVs to operate for longer periods without compromising payload capacity or mobility. Additionally, lithium-sulphur batteries offer an even higher energy density and have the added benefit of being more environmentally friendly, as sulphur is abundant and less toxic than the materials used in traditional batteries.

To address the power constraints faced by UGVs and UAVs in real-time coordination, energy harvesting technologies could be integrated into these platforms. For example, solar panels could be incorporated into UAVs to extend flight times during daylight hours [124], while UGVs could use kinetic energy recovery systems to recharge their batteries while in motion [125, 126]. These advancements would reduce the need for frequent recharging and enable more efficient deployment of unmanned systems in remote or hard-to-access areas.

Incorporating swarm robotics with UGVs and UAVs, enabled by advanced battery systems, could increase the overall mission coverage and reduce the risk of individual system failure due to energy depletion.

Table 9 shows the comparative analysis of several advanced battery technologies and their key characteristics in the context of UGV and UAV integration. The integration of advanced battery technologies is crucial for the future of UGV and UAV systems, as these technologies directly affect their operational efficiency, performance, and sustainability.

Energy density is a critical factor for both UGVs and UAVs, as higher energy density allows for longer operation times without significantly increasing weight [137, 138]. Table 9 shows that Lithium-Ion (Li-ion) and Solid-State Batteries stand out due to their high energy density, which directly translates to extended flight or operation times. UAVs face strict weight limitations, making high energy density a priority. Solid-state batteries outperform other technologies in terms of energy density, promising even longer operational durations, which is essential for missions that require extended range or endurance.

The charge/discharge cycles indicate how many times a battery can be charged and discharged before its performance significantly degrades [139]. The longevity of batteries is especially important for UGVs and UAVs, which often operate in harsh or remote environments where frequent battery replacements are impractical. Solid-state batteries again perform exceptionally well in this area, offering thousands of cycles, far exceeding the typical lifespan of Lithium-Ion or Lithium Polymer (LiPo) batteries. This means that Solid-State Batteries would be more cost-effective and reliable over time, reducing operational costs and maintenance needs for UGVs and UAVs.

The environmental impact of battery technology has become increasingly important as the demand for sustainable solutions rises. Many conventional batteries, such as Nickel-Metal Hydride (NiMH) and Lithium Polymer (LiPo), while effective, have moderate to high environmental impacts due to the mining processes and challenges in recycling their materials [140, 141]. On the other hand, sodium ion (Na-ion) batteries and Solid-State Batteries are considered more sustainable alternatives [142]. Sodium-ion batteries, for example, use more abundant and less environmentally damaging materials compared to Lithium-based batteries [143–145]. Solid-state batteries also hold promise in this area, as they tend to be more eco-friendly due to their potential for recyclable components and the elimination of harmful electrolytes [142]. This environmental consideration is crucial as the demand for eco-friendly solutions grows in industries focusing on automation, such as UGV and UAV deployment.

In conclusion, the adoption of advanced battery technologies such as Solid-State, Sodium-Ion, and Lithium-Ion represents the future direction for UGV-UAV integration. These technologies enable longer, more efficient operational periods, improve system sustainability through increased longevity and reduced environmental impacts, and reduce overall operational costs. As UGV and UAV applications continue to expand across various industries—such as logistics, defence, agriculture, and disaster response—advanced batteries will play a pivotal role in ensuring the long-term viability and effectiveness of these systems. Therefore, continued investment and research into these battery technologies are essential to unlock the full potential of autonomous and unmanned systems.

Table 9 Key features of advanced battery technologies

Battery technology	Energy density	Charge/Discharge cycles	Environmental impact
Lithium-ion (Li-ion)	High (150–250 Wh/kg) [127]	500–4,000 cycles [128]	Low environmental impact; recyclable
Lithium Polymer (LiPo)	Medium–High (100–200 Wh/kg) [127]	300–500 cycles [129]	Moderate impact; recycling challenges
Solid-State Batteries	Very High (500 + Wh/kg) [130]	8,000–10,000 cycles [131]	Low impact; potentially more eco-friendly
Sodium-Ion (Na-ion)	Medium (160–200 Wh/kg) [132]	1,000–6,000 cycles [133]	Lower environmental impact than Li-ion
Flow Batteries	Low (40–80 Wh/kg) [134]	15,000 + cycles [135]	Moderate impact: materials are more abundant
Nickel-Metal Hydride (NiMH)	Medium (70–100 Wh/kg) [134]	600 cycles [136]	Higher environmental impact than Li-ion

6 Results and Key Findings

Based on the comparative assessment using six core evaluation metrics, interoperability, communication, sensor fusion, autonomy, operational efficiency, and cybersecurity, the following key findings emerge:

1. Interoperability remains the most significant bottleneck in UGV-UAV integration. The proposed URP framework demonstrated the strongest potential to overcome cross-protocol barriers by unifying ROS and MAVLink and securing transmission using blockchain.
2. Real-time communication challenges such as latency and signal interference vary by environment. Hierarchical mesh networks supported by adaptive frequency management provide effective mitigation strategies in both urban and rural contexts.
3. Sensor fusion solutions that incorporate LiDAR, thermal, and optical sensors using AI-based algorithms show promise, though real-time multimodal integration continues to face processing limitations.
4. Autonomy & AI coordination, especially via swarm intelligence, improves task allocation and resilience in decentralized operations, particularly in disaster and military scenarios.
5. Operational efficiency is best enhanced through modular payload systems, which increase mission versatility, lower maintenance costs, and reduce system downtime.
6. Cybersecurity, particularly through blockchain and emerging QKD methods, is vital for mission integrity. These technologies offer secure data handling but vary in implementation complexity and cost.

The evidence supports the need for holistic, system-level frameworks that go beyond individual component optimization. These findings underscore the transformative potential of URP, AI-driven swarm coordination, and modular platforms in achieving effective UGV-UAV integration across real-world domains.

7 Conclusion

In conclusion, this review has explored the integration of UGVs and UAVs, focusing on both the challenges and potential solutions that can enhance the coordination of these systems for complex missions. The integration of UGVs and UAVs offers substantial advantages, such as increased operational flexibility, enhanced situational awareness, and improved mission efficiency. However, the successful deployment of these systems is impeded by a range of challenges, including communication issues, sensor incompatibility, and environmental factors. Addressing these challenges is crucial for realizing the full potential of UGV-UAV collaboration.

A central contribution of this review has been the introduction of the Unified Robotic Protocol (URP), a hybrid communication framework designed to bridge the gaps between the communication protocols used by UGVs and UAVs. By combining existing protocols like ROS and MAV Link, the URP offers a scalable solution to the interoperability problems that often arise in integrated systems. In addition, this review has emphasized the need for modular payload systems, which allow for dynamic adaptation to mission requirements, thus increasing the flexibility and effectiveness of UGV-UAV operations. The application of artificial intelligence and swarm intelligence in coordinating tasks among multiple agents further enhances system efficiency, enabling decentralized decision-making and autonomous operation under complex conditions.

Real-world case studies presented in this review demonstrate the practical applications of UGV-UAV integration, particularly in military operations, disaster response, and surveillance. These examples highlight the benefits of combined operations, such as faster decision-making, improved resource allocation, and enhanced operational safety. However, the case studies also underscore the importance of overcoming existing technical and operational constraints, including the need for secure communication systems and solutions to environmental limitations that affect system performance.

Looking forward, several emerging technologies offer promising solutions to these challenges. Blockchain technology holds the potential for ensuring secure and tamper-proof data exchanges between UGVs and UAVs, thereby enhancing the trustworthiness of autonomous systems. Similarly, quantum computing could play a pivotal role in addressing the complex optimization problems inherent in coordinating large-scale UGV-UAV operations. Moreover, advancements in battery technologies, such as solid-state and lithium-sulphur batteries, are crucial for extending the operational range and endurance of these systems, particularly in missions requiring prolonged deployment in remote or challenging environments.

While substantial progress has been made in the integration of UGVs and UAVs, further research is needed to refine the proposed solutions and explore new avenues for enhancing their performance. Future work should focus on developing standardized communication frameworks, improving sensor fusion techniques, and addressing the ethical and regulatory challenges that arise with the increasing autonomy of unmanned systems. As these technologies continue to evolve, the integration of UGVs and UAVs holds the potential to revolutionize mission strategies across a wide range of fields, from military operations to humanitarian aid and beyond.

The integration of UGVs and UAVs holds immense potential for enhancing operational efficiency, autonomy, and resilience across diverse applications. This review identified critical challenges, communication incompatibilities, sensor fusion difficulties, environmental constraints, and cybersecurity vulnerabilities, and assessed current solutions through a structured framework of interoperability, real-time communication, sensor accuracy, and AI integration. Emerging technologies such as the URP, blockchain security, modular payload systems, and swarm intelligence present promising pathways for overcoming these barriers. Real-world case studies in military, disaster response, and surveillance contexts highlight the complementary capabilities of UGVs and UAVs. As research progresses, integrating quantum computing, advanced encryption, and AI-driven coordination will be essential for fully realizing the potential of these autonomous systems. Future work should focus on developing standardized, secure, and scalable frameworks to enable seamless UGV-UAV collaboration in complex environments.

To further advance UGV-UAV integration, future research should focus on developing dynamic mission coordination algorithms that can adapt in real-time to unpredictable scenarios. These algorithms should combine elements of reinforcement learning, swarm intelligence, and metaheuristic optimization to allocate resources across heterogeneous platforms efficiently. Additionally, research should explore robust multi-agent simulation environments to test coordination frameworks under varied operational stressors such as signal loss, sensor failure, and adversarial interference.

Emerging fields such as neuromorphic computing and edge-AI also hold promise for improving real-time decision-making and reducing computational latency in UGV-UAV networks. Furthermore, interdisciplinary collaborations between AI researchers, roboticists, and systems engineers will be crucial in developing scalable and secure protocols, particularly for operations in contested or GPS-denied environments. Lastly, ethical and regulatory considerations, particularly regarding autonomous decision-making, data privacy, and accountability, must be more rigorously integrated into future development frameworks to ensure safe and responsible deployment.

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Data Availability The authors confirm that the data supporting the findings of this study are available within the article.

Code Availability Not applicable.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Conflicts of interest The authors declare no conflicts of interest to report regarding the present study.

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